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Photoelectrochemical cells for solar hydrogen production: current state of promising photoelectrodes, methods to improve their properties, and outlook

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Harnessing solar energy for the production of clean hydrogen fuels by a photoelectrochemical (PEC) cell represents a very attractive but challenging alternative. This review focuses on recent developments of some promising photoelectrode materials, such as $BiVO_4$, a-Fe_2O_3, TaON, and Ta_3N_5 for solar hydrogen production. Some strategies have been developed to improve PEC performances of the photoelectrode materials, including: (i) doping for enhancing visible light absorption in the wide bandgap semiconductor or promoting charge transport in the narrow bandgap semiconductor, respectively; (ii) surface treatment for removing segregation phase or surface states; (iii) electrocatalysts for decreasing the overpotentials; (iv) morphology control for enhancing the light absorption and shortening transfer distance of minority carriers; (v) other methods, such as sensitization, passivating layer, and band structure engineering using heterojunction structures, and so on. Photochemical durability of the photoelectrodes is also discussed, since any potential PEC technology must balance efficiency against cost and photochemical durability. Photochemical durability may be amended by optimizing the photoelectrode, electrocatalyst, and electrolyte at the same time. In addition, solar seawater splitting is briefly introduced because it has received attention recently. Finally, trends in research in PEC cells for solar hydrogen production are detailed.

Broader context

One essential issue facing humanity today is powering the Earth without producing additional CO_2 in the atmosphere. Solar energy can easily provide sufficient power for all of our energy demands if it can be efficiently harvested. Natural photosynthesis can store energy from sunlight in the chemical bonds of carbohydrates. Artificial photosynthetic routes, for example, photoelectrochemical solar energy conversion, can use solar energy to split water for hydrogen production, thus attracting more and more interest. However, the solar energy conversion efficiency is very low. This review assesses critically recent progress on some promising photoelectrodes with high potential efficiency, such as $BiVO_4$, a-Fe_2O_3, TaON, and Ta_3N_5, and discusses the approaches to improve their solar energy conversion efficiency and photoelectrochemical durability.

1 Introduction

Harvesting sunlight to produce clean hydrogen fuel remains one of the main challenges for solving the energy crisis and ameliorating global warming.[1–3] Photoelectrochemical (PEC) H_2 production, using solar energy to split water, is a promising method for providing clean energy carriers in the future. Since titanium dioxide (TiO_2) was reported to exhibit the ability for PEC water splitting, intensive and growing attention has been paid to photoelectrode materials for solar H_2 production, because they make use of the Earth's abundant, long lasting and clean solar energy.[4–8]

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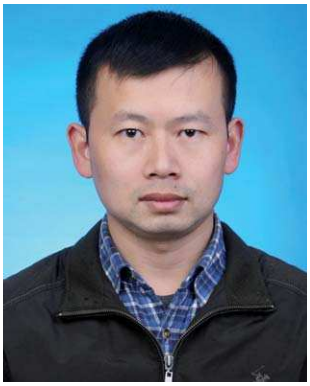
Early studies on photoelectrode materials for PEC cells have been focused on TiO_2. However, TiO_2 only absorbs ultraviolet light, which accounts for only 4% of the incoming solar energy on the Earth's surface, owing to its relatively large bandgap of about 3.0–3.2 eV. The PEC properties of TiO_2 photoelectrodes can be improved by tuning the morphology.[9–12] For example, Mor *et al.* reported that incident photon-to-electron conversion efficiency (IPCE) of a nano-array TiO_2 photoelectrode is up to 90% at 337 nm and 1.0 V *vs.* reversible hydrogen electrode (RHE).[9] Hydrogen treatment has also been used to upgrade fundamentally the performance of TiO_2 nanowires (*i.e.* H:TiO_2) for PEC water splitting.[13] However, owing to the large bandgap, only the ultraviolet part of the solar irradiation can be absorbed by TiO_2, and the theoretical solar-to-hydrogen efficiency (STH) of rutile TiO_2 cannot be over 2.2% (1.3% for anatase TiO_2) under air mass 1.5 global (AM 1.5 G) illumination. To enhance the STH, it is necessary to extend the light absorption edge of the photoelectrodes into the visible region, which possesses *ca.* 43% of the solar energy. Even though doping TiO_2-based photoelectrodes with metal cations or nonmetal anions may improve the IPCE in the visible region, this approach cannot bring an obvious increase in the STH.[14–19]

Compared with TiO_2, ZnO exhibits similar bandgap and band edge positions. Furthermore, ZnO has a direct bandgap and higher electron mobility than TiO_2, thus receiving considerable attention for PEC water splitting.[3,20–22] However, ZnO photoelectrodes usually exhibit less efficient PEC performances than TiO_2.[3]

Besides TiO_2 and ZnO, some oxide semiconductor photoelectrodes such as α-Fe_2O_3, WO_3, $SrTiO_3$, $KTaO_3$, and $BaTiO_3$ have been studied intensively.[3,6,23–27] Similar to TiO_2, $SrTiO_3$, $KTaO_3$, and $BaTiO_3$ suffer from low solar conversion performances due to their large bandgaps. WO_3 and α-Fe_2O_3 photoelectrodes have been studied the most intensively, on account of their narrow bandgaps as well as good stability and low price.[6,17,28,29] Augustynski and coworkers have reported that mesoporous WO_3 photoanodes exhibit the highest IPCE of *ca.* 80% at 400–410 nm, which is the maximum among the visible-light-responsive semiconductor photoelectrodes.[30] However, the theoretical STH of the WO_3 photoelectrode is only 4.8%. For a practical PEC water splitting system, at least 10% of STH appears to be necessary.[31] Since α-Fe_2O_3 exhibits 12.9% of the



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theoretical STH, α-Fe_2O_3 photoelectrodes have been a hot topic for several decades.[6,32]

Non-oxide semiconductor photoelectrodes with narrow bandgaps, such as Si, InP, GaP, $GaInP_2$, CdS, CdSe, CdTe, and amorphous silicon carbide (a-SiC:H), have also been studied for many years.[33–39] Unfortunately, most of them undergo photochemical corrosion under light irradiation.[37] However, highly purified planar crystalline Si provides promise as a photocathode for solar hydrogen production from water.[33,40–44] Arrays of B-doped p-Si microwires have yielded thermodynamically based energy conversion efficiencies >5% under 1 sun solar simulation.[21] Device-physics modeling has predicted that radial junction p–n Si microwire arrays can yield a solar energy-conversion efficiency of greater than 15%.[45]

Recently, many efforts have been made to develop new visible-light-responsive photoelectrodes, such as $BiVO_4$, $BiCu_2VO_6$, TaON, Ta_3N_5, $CuGa_3Se_5$, $(SrTiO_3)_{1-x}(LaTiO_2N)_x$, $CuWO_4$, and so on.[28–30,32,46–54] The theoretical STH values of $BiVO_4$, TaON and Ta_3N_5 are as high as 9.1%, 9.3% and 15.9%, respectively.[32] Therefore, these photoelectrode materials have attracted intensive and growing interest more recently.[55–60]

This review will focus on some progress on the study of these interesting and important photoelectrodes for solar hydrogen production, such as $BiVO_4$, α-Fe_2O_3, TaON and Ta_3N_5. The fundamental principles of PEC hydrogen production, measurement methods of IPCE, photoconversion efficiency and STH, will be summarised. This review will also summary some strategies, such as doping, surface treatment, electrocatalysts, morphology control, and so on, to optimize PEC efficiencies. Methods to improve the PEC durability of photoelectrode materials by optimization of photoelectrode, electrocatalyst, and electrolyte as well as surface protecting layer, will also be described. In addition, solar seawater splitting will be briefly discussed because it has received attention recently. Finally, we will discuss the study of PEC hydrogen production in the future.

2 Principle of PEC hydrogen production

Though the principle of PEC hydrogen production is relatively simple, the underlying fundamental processes are complex and still not well apprehended at the molecular level.[3] In general, a PEC cell consists of a working photoelectrode (for example TiO_2) and a counter electrode (for example a Pt cathode) immersed in an electrolyte. The structure of the semiconductor–electrolyte junction is comprised of a space charge layer of semiconductor photoelectrode, a Helmholtz layer (also named as the compact or inner layer), and a Gouy layer (also called the outer diffuse layer).[37,61–63] Fig. 1 shows the semiconductor–electrolyte junction structure of an n-type semiconductor. The thickness of the Helmholtz layer (II) is 0.3–0.5 nm, which is independent of the nature of the solid photoelectrode. However, the thicknesses of the space charge layer (Depletion layer, I) and Gouy layer (III) decrease with increasing concentrations of charge carriers in the photoelectrode and electrolyte, respectively. For example, the thickness range of the Gouy layer (III) is between 10 and 100 nm in aqueous ionic solutions with low ion concentrations, while it decreases to

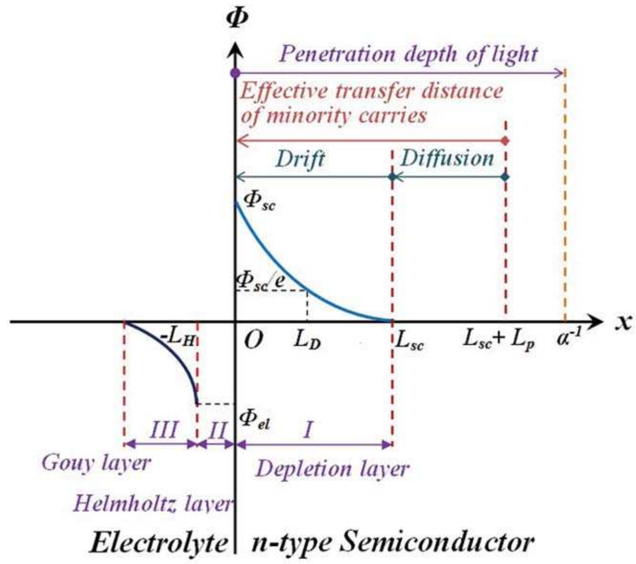

Fig. 1 Structure of a semiconductor–electrolyte junction. ϕ_{sc} is the potential drop within the depleted layer; L_D is the Debye length that can be regarded as the characteristic length of potential attenuation (ϕ_{sc}); ϕ_{el} is the potential drop within the Gouy layer; L_{SC} is the width of the space charge layer (the depletion layer is the most common and important type of the space charge layer); L_p is the minority carrier diffusion length and α is the optical absorption coefficient of the photoelectrode. For the depletion layer, $L_{sc} > L_D$, while $L_{sc} < L_D$ for accumulation layer or inversion layer. The number e is a mathematical constant, approximately equal to 2.71828.

about a mono-molecular layer in 1 M concentrated ionic solutions.[61] Accordingly, it can be omitted in PEC cells, because one often uses electrolytes with high ion concentrations.

For a semiconductor, the space charge layers can be classified into three types: (i) the *accumulation layer* in which mobile majority charge carriers are accumulated; (ii) the *depletion layer* in which mobile majority carriers are depleted and an excess space charge is carried by immobile ionized donors or acceptors; and (iii) the *inversion layer* in which mobile minority charge carriers are accumulated.[62] In general, the thickness of space charge layers (L_{SC}) can be written as,

$$L_{SC} = L_D\left[\frac{2q|\phi_{sc}|}{kT}\right]^{1/2} = L_D\left[\frac{2q(V - V_{fb})}{kT}\right]^{1/2} \quad (1)$$

$$L_D = \left[\frac{\varepsilon_0 \varepsilon kT}{q^2(n_0 + p_0)}\right]^{1/2} \quad (2)$$

where L_D depends on the concentration of charge carriers of a photoelectrode; ε_0 is the vacuum dielectric constant; ε is the relative dielectric constant for photoelectrode; k is the Boltzmann constant; T is in degrees Kelvin; V_{fb} is the flat potential of the photoelectrode; and n_0 and p_0 are the electron and hole concentration of the photoelectrode, respectively.[64] However, the depletion layer of a nanostructured photoelectrode, such as TiO_2 and WO_3, can be omitted, and the photogenerated carriers can be transferred only by diffusion.[65,66]

The IPCE of a photoelectrode, which is a vital parameter to describe its PEC performance, is mainly influenced by its optical absorption coefficient, the width of the space charge layer, and its minority carrier diffusion length (L_p).[62] The minority carrier diffusion length, which means the average distance that a minority carrier in a semiconductor can move from the site of generation until it is recombined, can be written as,

$$L_{\mathrm{p}} = \sqrt{D\tau} = \sqrt{\frac{\mu k_{\mathrm{B}}T}{q}\tau} \tag{3}$$

where D is the minority carrier diffusion coefficient and τ is the minority-carrier lifetime. The diffusion coefficient is related to the minority carrier mobility, μ ($m^2\ V^{-1}\ s^{-1}$). In the case of a minority-carrier, the carrier mobility can be written as,

$$\mu = \frac{q\tau}{m^*} \tag{4}$$

where m^* is the effective mass of minority-carrier.[61] During the diffusion of a minority carrier, the lifetime of the minority carrier (τ) may be influenced by donor levels. According to eqn (3) and (4), the decrease in m^* favors an increase the minority carrier diffusion length.

The photoinduced electron–hole pairs may be recombined in the bulk, the depletion layer, or through surface states, as shown in Fig. 2. In general, the recombination in the depletion layer is negligible. In doped semiconductors, deep energy levels, instead of shallow energy levels, are effective recombination centers. Therefore, shallow energy levels should be expected in doped photoelectrodes.[67–72] Surface states (also called *interfacial states*), which are two-dimensional localized levels associated with the semiconductor surface, can form due to abrupt distortion of the semiconductor crystal lattice or adsorbed ions. The former is an intrinsic surface state with short relaxation times (10^{-6} to 10^{-3} s), while the latter is an extrinsic surface state with long relaxation times (~1 s). Generally, the extrinsic surface state predominates over the intrinsic surface state.[61] In some cases, surface states may act in recombination of photogenerated electrons and holes, whereas they also can perform a transit shipment of the photogenerated holes for the oxidation reaction. Very recently, the role of surface states has been of much interest.[73–75]

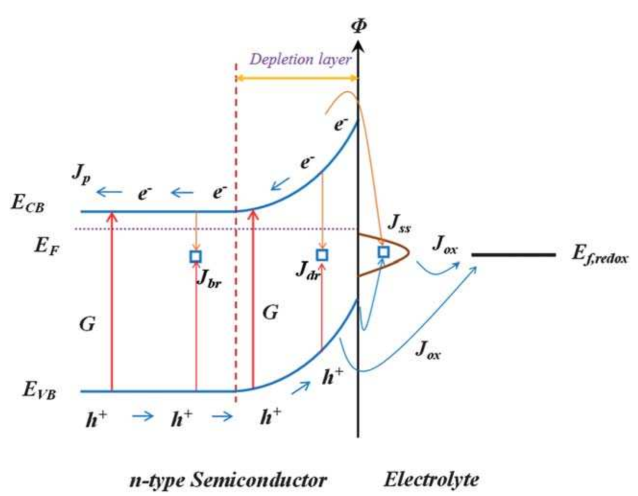

Fig. 2 Main recombination paths of photogenerated holes and electrons for an n-type semiconductor without any segregation. The unfavorable recombination pathways include bulk recombination (J_{br}), depletion region recombination (J_{dr}), and surface state recombination (J_{ss}). G is the favorable pathway of photogenerated electrons. Photogenerated holes can join directly in the oxidation reaction with the redox in the electrolyte, and the surface state also can act in a transit shipment of the photogenerated holes for the oxidation reaction.

The IPCE of a photoelectrode may be measured in two-electrode or three-electrode configurations. In the case of the two-electrode configuration consisting of a working photoelectrode and a counter electrode, the photocurrent density of the photoelectrode is influenced by the type of counter electrode, the distance between the two electrodes, and so on, thus possibly leading to an obvious error in the IPCE. In order to avoid this error, a reference electrode is introduced into the three-electrode configuration to exclude the effects of the counter electrode, which means that the reference electrode has no effect on the photocurrent of the photoelectrode, even with different counter electrodes or different distances from the photoelectrode. It is helpful to compare the measured IPCE with others' results at the same potential *vs.* RHE. Three-electrode configuration measurements are critical to interpret the measurements correctly.[76] Therefore, the IPCE of a photoelectrode measured by using a standard three-electrode configuration is an important parameter in evaluating its PEC activity. The IPCE of a photoelectrode at wavelength λ can be calculated as follows,[3]

$$\mathrm{IPCE}(\lambda) = \frac{1240 \times J_{\mathrm{p}}(\lambda)}{P \times \lambda} \times 100\% \tag{5}$$

where P and λ are the intensity ($\mu W\ cm^{-2}$) and the wavelength (nm) of incident light, respectively; $J_p(\lambda)$ is the photocurrent density ($\mu A\ cm^{-2}$) under the irradiation of single wavelength λ. One may integrate the observed IPCE data with the standard solar spectra (AM 1.5, 100 mW cm^{-2}) to verify the observed solar photocurrents.[32,48] For a photoelectrode, the plot of IPCE as a function of wavelength is known as the action spectra, which is often used in PEC studies.

The photoconversion efficiency (η) for a semiconductor photoelectrode is also a vital and essential parameter used to assess its PEC performances under solar irradiation. However, photoconversion efficiency of a photoelectrode is affected by the measurement system (a two-electrode or three-electrode configuration method), the light source, the counter electrode, the distance between the electrodes, the contact resistance, along with its band structure and reflection of radiation before reaching the photoelectrode.[32] The photoconversion efficiency for a photoelectrode can be estimated with the following equations,[77]

$$\eta = \frac{q\left(V_{\mathrm{ox}}^{0} - V_{\mathrm{app}}\right)}{I_0}\int_0^{1240/E_g} \mathrm{IPCE}(\lambda)N(\lambda)\mathrm{d}\lambda = \frac{J_{\mathrm{p}}\left(1.229 - V_{\mathrm{app}}\right)}{I_0},\ \text{for a photoanode} \tag{6a}$$

$$\eta = \frac{q\left(V_{\mathrm{app}} - V_{\mathrm{H_2}}^{0}\right)}{I_0}\int_0^{1240/E_g} \mathrm{IPCE}(\lambda)N(\lambda)\mathrm{d}\lambda = \frac{J_{\mathrm{p}}V_{\mathrm{app}}}{I_0},\ \text{for a photocathode} \tag{6b}$$

where q is the electron charge (1.6×10^{-19} C); E_g is bandgap of a photoelectrode; $N(\lambda)$ is the incident photon flux passed by the monochromator set to wavelength λ; V^0_{ox} and $V^0_{H_2}$ represent the standard potentials of oxygen and hydrogen electrodes, respectively; V^0_{ox} and $V^0_{H_2}$ take a value of +1.229 V and 0 V *vs.* normal hydrogen electrode in acidic electrolytes (pH = 0), respectively; I_0 is the light intensity of illumination (I_0 = 100 mW cm^{-2} if under AM 1.5 G solar illumination); J_p is the photocurrent density arising from PEC water oxidation or reduction; and V_{app} is the bias voltage applied between the working photoelectrode and counter electrode in the two-electrode configuration, while it is the bias voltage applied between the working photoelectrode and reference electrode in three-electrode configuration.

For the two-electrode configuration, similar to a solar cell, one can obtain the photoconversion efficiency of the photoelectrode (η), when the product of J_p and ($1.229 - V_{app}$) reaches a maximum. When the product of J_p and ($1.229 - V_{RHE}$) reaches a maximum, one can obtain the photoconversion efficiency of the photoelectrode (η) in a three-electrode configuration. For the three-electrode configuration, V_{app} should be converted into the potential *vs.* RHE. For example,

$$V_{app} = V_{Ag/AgCl} + 0.059\text{pH} + E^0_{Ag/AgCl}\ (E^0_{Ag/AgCl} = +0.199\text{V}),$$

when Ag/AgCl acts as a reference electrode. One should emphasize that the potential–photocurrent properties obtained by two- and three-electrode configuration methods with RHE correction are similar.[78]

Note that gaseous oxygen should be purged from the counter electrode compartment using a continuous flow of bubbled N_2 or Ar inert gas, because measurements performed in an oxygen-rich electrolyte can lead to erroneously large efficiency values.[32] One also should note that use of a xenon lamp as the light source can lead to a large overestimate of the solar-to-hydrogen efficiency, relative to that obtained under AM 1.5 G solar illumination.[32]

The solar-to-hydrogen efficiency (STH) for a PEC system constructed by photoelectrodes, where the only inputs are sunlight and water, can be expressed as,[79]

$$\text{STH} = \frac{J_p \times 1.229}{I_0} \times 100\% \qquad (7)$$

Theoretical STH and solar photocurrent of a photoelectrode under AM 1.5 G irradiation (100 mW cm^{-2}) are displayed in Fig. 3. It is based on an idealized case where 100% of the photons in the solar spectrum with energies exceeding the bandgap are absorbed and converted.[80] Although the thermodynamic limit implies that no photoelectrode with E_g < 1.229 eV displays properties for PEC water splitting, the practical limit, taking into consideration ohmic losses and a kinetic loss due to the overpotentials for oxygen or hydrogen production, is considered to *ca.* 2.03 eV, corresponding to a limiting bandgap wavelength of 610 nm.[32] Hypothetical ideal photoelectrode materials with a bandgap of 2.03 eV exhibit an STH of 16.8%.[32]

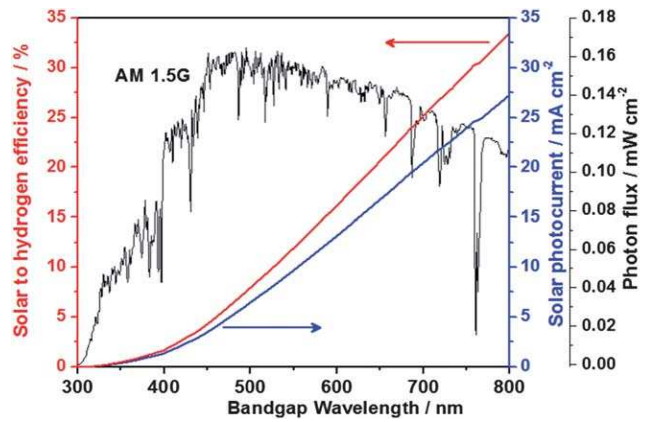

Fig. 3 Dependence of theoretical STH and solar photocurrent density of photoelectrodes on their bandgap absorption edges under AM 1.5 G irradiation (100 mW cm^{-2}).[32,79]

For a photoelectrode, J_p can be also described by eqn (8),

$$J_p = J_{abs} \times P_{sep} \times P_{inj} \qquad (8)$$

where J_{abs} is the photocurrent density originating from absorbed light with photoconversion efficiency of 100% (responding to the blue curve in Fig. 3); P_{sep} is the fraction of photogenerated holes that do not recombine with photogenerated electrons in the bulk and reach the photoelectrode–electrolyte interface; and P_{inj} is the yield of those holes that have reached the electrode–electrolyte interface and that are injected into the electrolyte to oxide the water, or, in other words, the holes that do not recombine with electrons at surface traps.[81] Obviously, P_{sep} and P_{inj} should be improved to increase PEC performance.

3 Current state of some promising photoelectrodes with high theoretical STH

Much attention has been paid to the development of oxides, especially $BiVO_4$ and hematite, as photoelectrode materials, because oxides are frequently inexpensive, stable, and easily prepared on a large scale. (Oxy)nitrides, especially TaON and Ta_3N_5, are also of much interest on account of their high theoretical STH.

3.1 Monoclinic scheelite $BiVO_4$

Among oxide photoelectrode candidates, monoclinic $BiVO_4$ has attracted broad interest as a water-oxidation photocatalyst under visible-light illumination.[82] As it is well known, $BiVO_4$ exists in three polymorphs of monoclinic scheelite, tetragonal scheelite, and tetragonal zircon structures, with bandgaps of 2.4, 2.34, and 2.9 eV, respectively. It is an attractive and nontoxic yellow pigment as a substitute for lead- and cadmium based yellow pigments. In 1998, monoclinic scheelite $BiVO_4$ was found by Kudo's group to exhibit very high activities for water oxidation under visible-light illumination.[82] The conduction band minimum and valence band maximum of $BiVO_4$ consists of V-3d and the hybridization between the O-2p and Bi-6s orbitals, respectively.[83–85] The visible light absorption is relative to the Bi 6s electron lone pairs.[82] The density functional theory (DFT) calculation results indicate that monoclinic $BiVO_4$ is a direct gap semiconductor. In theory, monoclinic $BiVO_4$ is a promising photoelectrode material with high PEC performances.[85,86] In 2003, Sayama *et al.* reported that the nanocrystalline $BiVO_4$ photoelectrode showed an IPCE of 29% at 420 nm and 1.9 V_{RHE} for PEC water splitting. Subsequent modification was carried out in 2006. Even though the IPCE of the nanostructured $BiVO_4$ photoanode at 420 nm and 1.6 V_{RHE} is only *ca.* 20%, it is a breakthrough for a multi-metal oxide photoanode.

However, the pure $BiVO_4$ photoelectrode is confronted with two difficulties. One is that its IPCE at lower potentials is very low, about 10% at 1.0 V_{RHE}, which is one of main problems that hinders its application in solar hydrogen production. A very high voltage must be applied to obtain a fairly high IPCE on $BiVO_4$ photoelectrodes, which may be related to the high surface recombination of photogenerated electrons and holes. Another problem is that the photocurrent stability of $BiVO_4$ is quite poor. Only 25% of the initial photocurrent was left after half an hour of illumination.[46] The photocurrent decrease comes from the V^{5+} dissolution into solution, and oxidation products on the surface of the photoelectrode, such as H_2O_2 or O_2, which play a role as recombination centers.

Since Sayama's report, monoclinic $BiVO_4$ has attracted more and more interest, owing to their high theoretical STH of 9.1%. Table 1 lists the PEC performances of $BiVO_4$-based photoelectrodes prepared with various methods.

3.1.1 Different preparation methods of $BiVO_4$ photoelectrodes. A suitable preparation method is crucial for high PEC performances of a photoelectrode material. The preparation method should be simple, cheap and easy to dope. The common preparation methods of $BiVO_4$ photoelectrodes include metal–organic decomposition,[46,55,87,97] powder spreading,[88,92] CBD,[93,94] USP,[95] reactive ballistic deposition,[96,100] electrodeposition,[106] electrophoretic deposition,[107] *etc.*

In 2008, Long *et al.* prepared $BiVO_4$ electrodes by powder spreading, in which a suspension of the $BiVO_4$ powder in absolute ethanol was spread onto the ITO (indium tin oxide) glass substrate, and then was sintered to improve adhesion.[88] However, the IPCE of the $BiVO_4$ electrode was very low and less than 2% at 420 nm (even under a very high potential of 1.6 V_{RHE}). The low IPCE may be related to not only the poor contact between the particles, but also bad crystallinity of the $BiVO_4$ caused by a low calcining temperature. The powder spreading method was improved by using fine $BiVO_4$ powders in 2010.[91] The IPCE was still low, about 12% at 440 nm and a very high potential of 2.1 V_{RHE}, possibly due to the poor contact and low carrier concentration.

Lee and coworkers have prepared PO_4-doped $BiVO_4$ photoelectrodes on FTO glass substrate using the EPD technique.[107] However, the PEC performance of the $BiVO_4$ samples was also limited by the poor contact between $BiVO_4$ particles.

A CBD method is cheap and easy to scale up for industrial production. $BiVO_4$ photoelectrodes have been prepared onto the FTO substrates, using the CBD method by the present authors.[93] $Bi(NO_3)_3 \cdot 5H_2O$ and NH_4VO_3 as Bi and V sources were dissolved in different aqueous solution, and ethylenediaminetetraacetic acid (EDTA) was added as a stabilizer. However, the IPCE of as-prepared $BiVO_4$ photoelectrodes is still low (*ca.* 5% at 400 nm and high potential 1.6 V_{RHE}), owing to large particle and poor contact between the particles. A $BiVO_4$ seed layer has been introduced in the CBD method to prepared $BiVO_4$ photoelectrodes on the FTO glass, thus obtaining the smaller particle size of $BiVO_4$, because the $BiVO_4$ seed layer favors its nucleation.[94] However, the PEC performance is still low, possibly due to its bad crystallinity.

The USP method has been introduced to prepare pure and W-doped $BiVO_4$ photoelectrodes by Guo and his coworkers.[95] The $BiVO_4$ electrode is less than 150 nm in thickness and highly transparent to wavelengths longer than 500 nm. The IPCE of $BiVO_4$ photoelectrodes at 420 nm and 0.9 V_{RHE} is only about 1% in 0.5 M Na_2SO_4 solution, not only because the $BiVO_4$ photoelectrode is too thin and cannot absorb more light, but also because the doping concentration is very low.

Berglund *et al.* have synthesized $BiVO_4$ films using a high vacuum chamber equipped with two electron beam evaporators.[96,100] Bismuth and vanadium metal were co-deposited on the FTO substrates and calcined at 500 °C for 2 h in air. For the $BiVO_4$ films deposited with excess vanadium, IPCE values above 21% and steady-state values above 14% were achieved in 0.5 M Na_2SO_4 at 1.6 V_{RHE} for light wavelengths of 340–460 nm.[96]

The electrodeposition procedure for $BiVO_4$ photoelectrodes has been developed by Seabold and Choi, using the nitric acid solution of $Bi(NO_3)_3$ and $VOSO_4$.[106] The bare $BiVO_4$ samples reach a maximum IPCE of 9% at 410 nm and 1.2 V_{RHE}. However, it is considerably difficult to prepare $BiVO_4$ photoelectrodes doped with elements.

Metal–organic decomposition is a facile method to synthesize $BiVO_4$ photoelectrodes, reported by various groups.[46,55,87,97] Up to date, the best performance of a $BiVO_4$ photoelectrode has been carried out by using metal–organic decomposition, which eases doping during the preparation. The samples prepared by this method possess good crystallinity and smaller grain size, though sintered under moderate high temperature.

3.1.2 $BiVO_4$ photoanode doped with elements. Nonisovalent substitutional doping often increases the carrier concentration of $BiVO_4$. Using 3% Mo doping into the site of V in $BiVO_4$, the IPCE of $BiVO_4$ can be increased from 9% to 41% at 420 nm and 1.0 V_{RHE} in our lab.[55,60] In order to comprehend the effects of nonisovalent substitutional doping on the PEC properties, various ions (Mo^{6+}, W^{6+}, Ta^{5+}, Zr^{4+}, Si^{4+}, Ti^{4+}, La^{3+}, Fe^{3+}, Sr^{2+}, Zn^{2+} and Ag^{+}) were doped into the sites of V in $BiVO_4$. The results indicated that only doping with Mo^{6+} or W^{6+} obviously enhanced the PEC performances. When Mo^{6+} substitutes for V^{5+}, it acts as an electron donor in the $BiVO_4$ lattice. The carrier concentration of 3%Mo-doped $BiVO_4$ photoelectrodes increases by 80 times, compared with pure $BiVO_4$ photoelectrodes. High conductivity can promote the photo-generated electrons to transfer from the thin film onto the substrate, which in turn improve the PEC performances.

Guo *et al.* have also reported 1% W-doping in $BiVO_4$ can increase the IPCE from 0.9% to 10% at 0.3 V_{SCE}, because the carrier concentration increases significantly after the W

Table 1 PEC performances of $BiVO_4$-based photoelectrodes[a]

Photoelectrode	Preparation	Catalyst	Electrolyte	Photoresponse	Ref.
$BiVO_4$	MOD	—	0.5 M Na_2SO_4	IPCE *ca*. 33% at 400 nm and 29% at 420 nm (1.9 V_{RHE})	46
$BiVO_4$	MOD	—	0.5 M Na_2SO_4	IPCE *ca.* 20% at 420 nm and 1.6 V_{RHE}	87
$BiVO_4$	MOD	Ag ion treatment	0.5 M Na_2SO_4	IPCE *ca.* 44% at 420 nm and 1.6 V_{RHE}	87
$BiVO_4$	Powder spreading	—	0.5 M Na_2SO_4	IPCE 1.5% at 420 nm and 1.6 V_{RHE}	88
$BiVO_4$	Powder spreading	Co_3O_4	0.5 M Na_2SO_4	IPCE 6.5% at 420 nm and 1.6 V_{RHE}	88
$BiVO_4$	MOD	—	0.5 M Na_2SO_4	0.173 μA at 1.1 V_{RHE}[b]	89
$BiVO_4$	MOD	—	Saturated $NaHCO_3$ with CO_2 gas bubbling	IPCE 32% at 420 nm and 1.23 V_{RHE} 45% at 420 nm and 1.9 V_{RHE}	90
$BiVO_4$	Powder spreading	—	0.1 M K_2SO_4	IPCE 12% at 440 nm and 1.5 V *vs.* Ag/AgCl.	91
$BiVO_4$/RGO	Powder spreading	—	0.1 M Na_2SO_4	IPCE 4.2% at 400 nm and 0.75 V *vs.* Ag/AgCl	92
$BiVO_4$	CBD	—	0.5 M Na_2SO_4	1.4 mA cm^{-2} at 1.6 V_{RHE}[c]	93
$BiVO_4$	CBD	—	0.5 M Na_2SO_4	0.4 mA cm^{-2} under AM 1.5 (1.6 V_{RHE})	94
$BiVO_4$	USP	—	0.5 M Na_2SO_4	IPCE ∼1% at 420 nm and 0.3 V_{SCE}	95
1%W–$BiVO_4$	USP	—	0.5 M Na_2SO_4	IPCE ∼10% at 400–450 nm and 0.3 V_{SCE}	95
$BiVO_4$	Reactive ballistic deposition	—	0.5 M Na_2SO_4	IPCE over 21% at 340–460 nm and 1.0 V *vs.* Ag/AgCl (1.6 V_{RHE})	96
7%W–$BiVO_4$	MOD	Co-Pi	0.1 M K_3PO_4 buffered to pH = 8	IPCE ∼33% at 370 nm and 1.23 V_{RHE}	97
$BiVO_4$	Spray pyrolysis	—	0.15 M K_2SO_4 (pH 7)	IPCE ∼6% at 450 nm and 1.63 V_{RHE}	98
FTO/SnO_2/$BiVO_4$	Spray pyrolysis	—	0.15 M K_2SO_4 (pH 7)	IPCE ∼25% at 450 nm and 1.63 V_{RHE}	98
FTO/SnO_2/1%W–$BiVO_4$	Spray pyrolysis	—	0.15 M K_2SO_4 (pH 7)	IPCE 46% at 450 nm and 1.63 V_{RHE}	98
3%Mo–$BiVO_4$	MOD	RhO_2	0.5 M K_2SO_4	3.3 mA cm^{-2} under AM 1.5 G (1.6 V_{RHE})	55
3%Mo–$BiVO_4$	MOD	—	Seawater	IPCE 41% at 420 nm and 1 V_{RHE}	55
3%Mo–$BiVO_4$	MOD	RhO_2	Seawater	IPCE 49% at 420 nm and 1 V_{RHE}	55
W–$BiVO_4$ (Bi/V/W = 4.5/5/0.5)	Deposition-annealing	—	0.1 M Na_2SO_4 with 0.1 M Na_2SO_3	IPCE *ca.*58% at 400 nm and 0.3 V *vs.* Ag/AgCl	99
W/Mo–$BiVO_4$ (Bi/V/W/Mo = 4.6/4.6/0.2/0.6)	Deposition-annealing	—	0.1 M Na_2SO_4	IPCE *ca.* 8% at 400 nm and 0.6 V *vs.* Ag/AgCl	100
2%Mo–$BiVO_4$	MOD	Co-Pi	0.5 M Na_2SO_4 (phosphate buffer at pH = 7)	IPCE 35% at 450 nm and 0.9 V *vs.* Ag/AgCl. (1.5 V_{RHE})	101
FTO/SnO_2/$BiVO_4$	Spray pyrolysis	Co-Pi	0.5 M K_2SO_4 buffered to pH ∼5.6 with 0.09 M KH_2PO_4/ 0.01 M K_2HPO_4	1.7 mA cm^{-2} under AM 1.5 G (1.23 V_{RHE})	102
WO_3/$BiVO_4$	MOD	—	0.5 M Na_2SO_4	IPCE 23% at 420 nm and 1.0 V *vs.* Ag/AgCl (1.6 V_{RHE})	103
WO_3/$BiVO_4$	Deposition-annealing	—	0.5 M Na_2SO_4	IPCE 38% at 420 nm and 0.7 V *vs.* Ag/AgCl (1.3 V_{RHE})	104
WO_3/$BiVO_4$	CBD	—	0.5 M Na_2SO_4	IPCE 31% at 420 nm and a bias of 0.5 V (two-electrode setup); 1.6 mA cm^{-2} at 1 V bias, backside AM 1.5 G illumination (two-electrode setup)	105
$BiVO_4$	Electro-deposition	FeOOH	0.1 M KH_2PO_4	IPCE 56% at 410 nm and 1.2 V_{RHE}	106

Table 1 (*Contd.*)

Photoelectrode	Preparation	Catalyst	Electrolyte	Photoresponse	Ref.
PO_4-doped $BiVO_4$	EPD	—	0.5 M Na_2SO_4	0.548 mA cm^{-2} at 1.23 V_{RHE} (300 W Xe lamp, $\lambda \geq 420$ nm)[b]	107
$BiVO_4/WO_3$	MOD	—	0.1 M $KHCO_3$ with CO_2 bubbling	IPCE *ca.* 31% at 420 nm and 1.2 V_{RHE}	78
$BiVO_4/SnO_2/WO_3$	MOD	—	0.1 M $KHCO_3$ with CO_2 bubbling	IPCE *ca.* 43% at 420 nm and 1.2 V_{RHE} 2.82 mA cm^{-2} at 1.23 V_{RHE} (AM 1.5)	78
$BiVO_4$	EPD	Co-Pi, CoO_x, IrO_x, MnO_x	0.1 M K_3PO_4 containing 0.8 g $NaIO_3$ (0.02 M, 200 mL) pH 7.0	Photocurrent density CoPi/$BiVO_4$ > $CoO_x/BiVO_4$ > $IrO_x/BiVO_4$ > $BiVO_4$ > $MnO_x/BiVO_4$	108
$FTO/SnO_2/BiVO_4$	MOD	—	0.5 M Na_2SO_4	IPCE 6.2% at 350 nm and 1.6 V_{RHE}	109

[a] Metal–organic decomposition (MOD); electrophoretic deposition (EPD); chemical bath deposition (CBD); reducing graphene oxide (RGO); ultrasonic spray pyrolysis (USP); fluorine-doped tin oxide (FTO); saturated calomel electrode (SCE). Unit of doping concentration is at. %. [b] The area of photoelectrode was not given. [c] The power of Xe lamp is 500 W. IPCE was calculated to be *ca.* 5% at 400 nm and 1.6 V_{RHE} by the present authors.

doping.[95] On account of the low W-doping concentration and the thin thickness of 150 nm, which cannot absorb as much light, the IPCE of W-doping $BiVO_4$ photoelectrodes may be expected to improve obviously by optimizing the doping concentration and the photoelectrode thickness.

To improve the PEC performance of $BiVO_4$, Bard's group has used similar strategy of nonisovalent substitutional doping (such as W, Fe, B, Cu, Zn, Ti, Nb, Sn, Co, Pd, Rb, Ru, Ag, Ga, Sr, and Ir).[99] Among them, W doping brings the most noticeable enhancement in the photocurrent. Compared with pure $BiVO_4$ photoelectrodes, the carrier concentration of the W-doped $BiVO_4$ photoelectrode increases twice, and their IPCE values are increased from 18% to ~58% at 0.3 V *vs.* Ag/AgCl in Na_2SO_4 solution with Na_2SO_3 as a hole sacrificial reagent. If there were no Na_2SO_3, the IPCE will be much lower, about 3.5%, which may come from the high W concentration of 10% and no suitable surface pretreatment. Codoping W and Mo into $BiVO_4$ photoelectrodes has improved electron–hole separation by significantly changing the bandgap or optical properties.[100]

Density functional theory studies have shown that Mo is better than W for the n-type doping of $BiVO_4$.[79,104] Mo doping into $BiVO_4$ increases the carrier concentration and distorts the VO_4 tetrahedron, thus forming a local internal electric field.[85,86] Mo-doping $BiVO_4$ exhibits excellent n-type conductivity and could significantly enhance the PEC performances.

3.1.3 $BiVO_4$ photoanode loaded electrocatalysts. Photoelectrochemical performances of $BiVO_4$ photoelectrodes are often limited by two main factors: slow water oxidation kinetics and poor charge separation.[102] Electrocatalysts, such as Co_3O_4, RhO_2, and Co-Pi, are also loaded onto the surface of $BiVO_4$ photoanodes to decrease the bias potential and improve the PEC stability.[55,88,102,108] Long and coworkers have reported that the IPCE of $BiVO_4$ in 0.5 M Na_2SO_4 solution increased from 1.5% to 6.5% at 420 nm and 1.6 V_{RHE} after Co_3O_4 surface modification, owing to the formation of an n-$BiVO_4$–p-Co_3O_4 junction.[88] However, they have ignored the effect of the external bias on the relative band positions. It is reasonable that Co_3O_4 acts as a water oxidation catalyst (WOC).

Mo-doping $BiVO_4$ photoelectrodes have been modified by RhO_2 for solar seawater splitting.[55] The IPCE increases from 41% to 49% at 420 nm and 1 V_{RHE}, since RhO_2 is considered to accelerate the photo-generated hole transfer rate. Also, the RhO_2 catalyst further improves significantly the photocurrent stability of the photoelectrodes. To date, the IPCE values of the $BiVO_4$ photoelectrodes at 440–480 nm have been the highest among all stable oxide photoanodes at this potential. The onset potential of the $BiVO_4$ photoelectrodes shifts cathodically about 120 mV.

Cobalt-phosphate (Co-Pi) is an efficient WOC for pure $BiVO_4$, and an AM 1.5 photocurrent of 1.7 mA cm^{-2} at 1.23 V_{RHE} has been obtained by Abdi and Krol.[102] Gamelin's group have reported that modification of W-doped $BiVO_4$ photoanode surfaces with Co-Pi has yielded a very large cathodic shift of ~440 mV in the onset potential for sustained PEC water oxidation at pH = 8.[97] The low absolute onset potential of ~0.31 V_{RHE} achieved with the Co-Pi/W-doped $BiVO_4$ combination is promising for overall solar water splitting in low-cost tandem PEC cells, and is encouraging for application of this surface modification strategy to other candidate photoanodes.[97]

In an attempt to mend the water oxidation kinetics of the $BiVO_4$ photoelectrodes, a layer of FeOOH was placed on the $BiVO_4$ surface as an oxygen evolution catalyst.[106] The IPCE of bare $BiVO_4$ photoelectrodes reaches a maximum of 9% at 350 nm and 1.2 V_{RHE}, while that of $BiVO_4$/FeOOH achieves 44% at this wavelength.[106] The $BiVO_4$/FeOOH films are capable of utilizing a substantially greater number of holes that reached the semiconductor–liquid junction for photo-oxidation of water.[106]

3.1.4 $BiVO_4$-based composite photoanode. The IPCE of $BiVO_4$ is increased remarkably from ~1.6% to 6.2% at 350 nm and 1.6 V_{RHE} when an SnO_2 film is layered between the $BiVO_4$ layer and the FTO substrate ($FTO/SnO_2/BiVO_4$), since the SnO_2

inner (or buffer) layer can assist the efficient electron–hole separation *via* the electron transfer from $BiVO_4$ to SnO_2, and the blocking of the hole transfer from $BiVO_4$ to the FTO surface.[109] Similar to SnO_2, WO_3 also acts as a buffer layer for $BiVO_4$ photoelectrodes.[103] Chatchai *et al.* have reported that the IPCE of the $BiVO_4$ composite photoelectrodes increases from *ca.* 7.8% (FTO/$BiVO_4$) to 82% (FTO/WO_3/$BiVO_4$) at 350 nm and 1.6 V_{RHE} by using the WO_3 buffer layer, when illuminated from the front side.[103] In this case, the WO_3 buffer layer can enlarge the electron–hole separation and suppress the recombination effect to enable the high PEC performances of the composite photoelectrode.[103] Significant enhancement in IPCE of WO_3–$BiVO_4$ heterojunction photoelectrodes (FTO/WO_3/$BiVO_4$) has also been found by Hong *et al.*, because the composite photoelectrode combines the merits of the two semiconductors, *i.e.* the excellent charge transport characteristics of WO_3 and the good light absorption capability of $BiVO_4$.[104]

The heterojunction films of WO_3–$BiVO_4$ photoanodes have been prepared by solvothermal deposition of a WO_3 nanorod-array film onto FTO glass, with the subsequent deposition of a $BiVO_4$ layer by spin coating. Compared with planar WO_3–$BiVO_4$ heterojunction films, the nanorod-array films show significantly improved PEC properties owing to the large specific surface area and improved separation of the photogenerated charges at the WO_3–$BiVO_4$ interface.[105]

As mentioned above, it is interesting that doping with W can remarkably improve the IPCE of $BiVO_4$ photoelectrodes. Note that the interface ions diffusion between WO_3 and $BiVO_4$ may take place when calcined at high temperature. It is unclear if the photocurrent enhancement of the composite photoelectrode comes from the heterojunction or doping with W.

3.2 Hematite (α-Fe_2O_3)

Hematite is envisioned as one of the promising photoelectrodes for solar hydrogen production, on account of its good photochemical stability, abundance, non-toxicity, and narrow bandgap ($E_g \sim 2.0$ eV).[6,7,110,111] However, PEC performances of hematite have been severely restricted by its shorter hole diffusion length (2–4 nm) than TiO_2 (on the order of 1 μm) and WO_3 (on the order of 100 nm), owing to the ultrafast recombination of photogenerated carriers (time constants on the order of 10 ps), along with its lower minority charge mobility (0.2 cm^2 V^{-1} s^{-1}).[112–116]

The conduction band minimum of hematite consists of Fe-3d e_g, while the valence band maximum is composed of Fe-3d t_{2g} (the main part) and O-2p.[117] To be specific, the source of photogenerated holes can be divided into two parts; one is from the direct O^{2-} 2p → Fe^{3+} 3d transitions (for wavelengths $\lambda < 400$ nm), and the other comes from the indirect Fe^{3+} 3d → 3d transitions ($\lambda < 620$ nm).[118] The direct transition results in a larger absorption coefficient compared with the indirect transitions, and thereby the action spectrum of hematite displays a precipitous decline over wavelength. It is a challenge to make the shape of the action spectrum flat in the visible region. There are some good review articles about hematite.[6,17,79] In this review, we discuss only the effects of deposition methods, doping, electrocatalysts, and passivating layers on PEC properties of hematite.

3.2.1 Deposition methods of hematite photoelectrode. The PEC performances of hematite photoelectrodes are very sensitive to the deposition methods. The state of the art hematite photoelectrodes have been prepared by the atmospheric pressure chemical vapor deposition (APCVD).[29] USP is also an effective method of fabricating hematite photoelectrodes with superior performances.[119,120] Grätzel *et al.* have used USP and APCVD to deposit a leaflet-type and a fractal-like cauliflower nanostructured hematite photoanode, respectively.[29,48,101,119,121] The special nanostructures are useful in efficient harvesting visible light and reducing the average distance that the photogenerated hole has to travel toward the electrolyte–electrode interface, thus resulting in lower recombination and favoring enhancement in PEC performances. The IPCE of the hematite photoelectrodes prepared by USP is 16% with 375 nm illumination at 1.2 V_{RHE}, and over 30% at 1.6 V_{RHE}.[119] In our experiments, the hematite photoelectrodes prepared by USP have exhibited no leaflet types or other special morphologies, but they do exhibit an accumulation of very fine particles, which also give relatively high PEC performances.[120] Grätzel *et al.* have reported that the solar photocurrent of hematite photoelectrodes can reach 3.3 mA cm^{-2} at 1.43 V_{RHE}, corresponding to STH = 4.1%.[29]

In some sense, USP is a better method than APCVD, though the state of the art hematite photoanodes are made by APCVD. The iron precursor used in APCVD is iron pentacarbonyl, which is toxic. Though ferrocene has been used in replacing iron pentacarbonyl, the solar photocurrent obtained remains less than 1 mA cm^{-2}.[122] Furthermore, the equipment of APCVD is relatively more expensive and complex than that of USP.[123] So if hematite photoelectrodes could be made by USP with photocurrents as high as those made by APCVD, it would be attractive.

Other approaches, such as hydrothermal and sonoelectrochemical anodization methods, have also been endeavored to synthesize various nanostructures (nanorod, nanotube, and nanowire) of hematite.[124–126] Recently, Li *et al.* have reported a facile method for preparing highly photoactive hematite films by a new deposition annealing process using nontoxic $FeCl_3$ as the Fe precursor, followed by annealing at 550 °C in air, as well as their implementation as photoelectrodes for PEC water splitting.[127] Very recently, atomic layer deposition (ALD) has been used to prepare Mg-doped hematite, which favors a uniform distribution of dopant.[128]

3.2.2 Hematite photoelectrodes doped with elements. Since pure hematite is an insulator, poor conductivity of hematite limits its PEC properties.[7,129,130] Owing to the precursor used in the deposition process, such as $Fe(acac)_3$ or $FeCl_3$, carbon or chlorine residues may remain in the hematite films and act as electron donors, thus yielding hematite films unintentionally doped with ions, and their resulting appreciable photocurrents.[131] In most cases, doping is done intentionally to improve hematite conductivity and photoresponses. Table 2 lists the current research results for doped hematite photoelectrodes and their corresponding photoresponses.

The effects of substitutional doping with elements, such as W, Mo, V, Nb, Ta, Si, Ti, Ge, Zr, Ru, Sn, Ce, Hf, Pt, Mg, Ca, Ni, Cu, Zn, and Cd (as listed in Table 2), on the PEC performances of hematite have been studied intensively since the 1970s. Nonisovalent substitutional doping (as listed in Table 2) can improve the carrier concentration and electrical conductivity of hematite. At the same time, the increased carrier concentration can reduce L_{sc}, which is deleterious for photogenerated holes' contribution to the photocurrent, based on the Schottky model.[116] However, the hematite hole lifetime is very short, so a narrower L_{sc} that shortens the hole transit time may be beneficial for charge separation under this condition.[155] For example, for Sn-doped hematite nanostructure, ultrafast spectroscopy studies revealed that there is significant electron–hole recombination within the first few picoseconds, while Sn doping and the change of surface morphology have no major effect on the ultrafast dynamics of the charge carriers on the picosecond time scales.[26]

Among these nonisovalent substitutional doping elements, the dopants Si^{4+} and Ti^{4+}, have been studied most intensively. It

Table 2 Photoresponses of some doped hematite photoelectrodes[a]

Dopant	Concentration	Technique	Photoresponses	Ref.
W	0.0025%	Synthesized ceramics	IPCE 29% at 400 nm and 1.23 V_{RHE}	132
Mo	15%	Electrodeposition	IPCE 12% at 400 nm and 1.4 V_{RHE}	133
V	0.05%	Synthesized ceramics	IPCE 28% at 400 nm and 1.23 V_{RHE}	132
Nb	0.5%	Synthesized ceramics	IPCE 27% at 400 nm and 1.23 V_{RHE}	132
Nb	0.1%	Synthesized ceramics	135 cm^{-2} at 0.2 V_{SCE} (100 W Hg arc lamp, 20 mW cm^{-2})	134
Nb	1.5%	Single crystal	IPCE 37% at 370 nm and 1.23 V_{RHE}	135
Nb	10%	Synthesized ceramics	IPCE 26.7% at 450 nm and 1.5 V_{RHE}	136
Ta	0.5%	Synthesized ceramics	IPCE 32% at 400 nm and 1.23 V_{RHE}	132
Ta	0.5%	Synthesized ceramics	640 μA cm^{-2} at 0.7 V (a high-pressure lamp of 1 KW) (two-electrode setup)	137
Si	2%	Synthesized ceramics	IPCE 34% at 400 nm and 1.23 V_{RHE}	132
Si	2%	Synthesized ceramics	IPCE 9.3% at 340 nm and 1.49 V_{RHE}	138
Si	~1.5%	APCVD	IPCE 42% at 370 nm and 1.23 V_{RHE}	131
Si	5.3%	RMS	IPCE 2.5% at 375 nm and 1.54 V_{RHE}	139
Si	0.1%	SP	IPCE 23% at 360 nm and 1.4 V_{RHE}	140
Si	0.2%	SP	0.37 mA cm^{-2} at 1.23 V_{RHE} (0.8 sun, AM 1.5)	140
Ti	5%	RMS	IPCE 14% at 375 nm and 1.54 V_{RHE}	139
Ti	4%	RBD	IPCE 31% at 360 nm and 1.4 V_{RHE}	141
Ti	2.5%	PVD	IPCE 15% at 360 nm and 1.6 V_{RHE}	142
Ti	1%	Synthesized ceramics	IPCE 1.2% at 340 nm and 1.49 V_{RHE}	138
Ge	0.05%	Synthesized ceramics	IPCE 5.5% at 340 nm and 1.44 V_{RHE}	138
Zr	0.5%	Synthesized ceramics	IPCE 23% at 400 nm and 1.23 V_{RHE}	132
Zr	2.0%	Electrodeposition	2.1 mA cm^{-2} at 1.64 V_{RHE} (Xe lamp, 150 mW cm^{-2})	143
Ru	5%	Synthesized ceramics	108 μA cm^{-2} at 0.2 V_{SCE} (100 W Hg arc, 20 mW cm^{-2})	134
Sn	4%	RBD	IPCE 21% at 360 nm and 1.4 V_{RHE}	141
Sn	0.05%	Synthesized ceramics	IPCE 4.0% at 340 nm and 1.04 V_{RHE}	138
Sn	4%	Deposition annealing	IPCE 0.7% at 420 nm and 0.3 V *vs.* Ag/AgCl (0.2 M NaOH)	144
Sn	9.4%	Hydrothermal method	IPCE 19.2% at 400 nm and 1.23 V_{RHE} 1.86 mA cm^{-2} at 1.23 V_{RHE} (AM 1.5 G)	26
Ce	0.05%	Synthesized ceramics	IPCE 30% at 400 nm and 1.23 V_{RHE}	132
Hf	0.002%	Synthesized ceramics	IPCE 26% at 400 nm and 1.23 V_{RHE}	132
Pt	5%	Electrodeposition	IPCE ~12% at 400 nm and 1.4 V_{RHE}	145
Pt	5%	Hydrothermal	1.38 mA cm^{-2} at 1.25 V_{RHE} under 1 sun illumination	146
Al	0.46%	Electrodeposition	IPCE 8% at 400 nm and 1.4 V_{RHE}	147
Cr	5%	Electrodeposition	IPCE 6% at 400 nm and 1.4 V_{RHE}	133
In	8–20%	Sputtering	21% at 400 nm and 1.5 V_{RHE} (effective quantum efficiency)	148
Mg	1%	Plasma sprayed	2.1% at 400 nm and −0.4 V (cathodic) *vs.* V_{SCE} (pH 13) 0.84% at 400 nm and +0.5 V (anodic) *vs.* V_{SCE} (pH 13)	149
Ca	1%	Synthesized ceramics	—	150
Ni	5%	Spray pyrolysis	0.26 mA cm^{-2} at 0.59 V_{NHE} (500 W Xe)	151
Cu	5%	Spray pyrolysis	66.8 μA cm^{-2} at 1.61 V_{RHE} (150 W Xe lamp)	152
Zn	5%	Spray pyrolysis	422.6 μA cm^{-2} at 1.61 V_{RHE} (150 W Xe lamp)	152
Zn	5%	Spray pyrolysis	0.64 mA cm^{-2} at 1.71 V_{RHE} (150 W Xe lamp)	153
Cd[b]	~1%	Electrodeposition	1.25 mA cm^{-2} at 1.23 V_{RHE} (AM 1.5)	154
Si + Ti	4%Si + 2%Ti	USP	IPCE 37% at 370 nm and 1.23 V_{RHE}	120
Be + Sn	6%Be+ 4%Sn	Deposition annealing	IPCE 1.2% at 420 nm and 0.3 V *vs.* Ag/AgCl (0.2 M NaOH)	144
Ti + Al	5%Ti + 1%Al	Spray pyrolysis	IPCE 25% at 400 nm and 0.7 V_{NHE}	151
Ti + Li	5%Ti + 1%Li	Spray pyrolysis	0.9 mA cm^{-2} at 0.5 V_{NHE} (500 W Xe)	151

[a] Note: RMS (reactive magnetron sputtering), RBD (reactive ballistic deposition), SP (spray pyrolysis), PVD (physical vapor deposition). Unit of doping concentration is at. %. [b] Cd exists mainly as CdO and/or $Cd(OH)_2$ in the hematite surface.

is still a debate whether the increase in conductivity of Ti doped hematite is a result of the dopant acting as a donor, or small polaron hopping (Fe^{3+} + Ti^{4+} → Fe^{2+} + Ti^{3+}).[17] Apart from Si^{4+} and Ti^{4+}, other tetravalent cations, like Sn^{4+}, Zr^{4+}, and Ge^{4+}, have also been incorporated into hematite. Carter *et al.* have concluded that zirconium, silicon, or germanium doping is superior to titanium doping because the former dopants do not act as electron trapping sites, by reason of the higher instability of Zr^{3+}(III) compared to Ti^{3+}(III) and the more covalent interactions between silicon (germanium) and oxygen.[156] However, the results of the present authors[120] and Glasscock *et al.*[139] have indicated that titanium is a better dopant than silicon in improving hematite photoresponses. To further increase the carrier concentration, we have proposed an idea of codoping with Si^{4+} and Ti^{4+}. The results show that codoping is superior to Si-doping or Ti-doping, and it significantly enhances the hematite photoresponses. The reason that codoping significantly elevates the donor concentration may be the ion radii balance between Si^{4+}, Fe^{3+} and Ti^{4+}.[120] The IPCE at 400 nm is the highest among the a-Fe_2O_3 samples prepared by USP.

Whereas substitutional doping with hexavalent, pentavalent, and tetravalent metal ions is generally considered to increase donor levels, divalent or monovalent substitutional doping, such as Mg^{2+}, Ca^{2+}, Cu^{2+}, is used to obtain p-type semiconductors (photocathodes) by increasing positive charge carriers. However, this p-type behavior was not repeated by some groups,[17] which may be ascribed to the presence of oxygen defects in hematite photoelectrodes.

Simultaneous doping with dopants that should favor the formation of n- and p-type hematite, for example codoping with Ti and Li, has been also studied by Augustynski and coworkers.[151] The photocurrent responses of the hematite doping with Ti and Li are only improved slightly.

Besides nonisovalent substitutional doping, isovalent substitutional doping, like Al^{3+}, In^{3+}, and Cr^{3+}, have also been investigated.[133,147,148,151] Adding small quantities of dopants which should, in principle, form oxides that are isostructural with hematite, thereby directing the crystallinity of the hematite photoelectrode.[151] Unlike nonisovalent substitutional doping which creates excess electrons–holes in the conduction process, Al^{3+} isovalent substitutional doping causes a contraction of the crystal lattice due to the smaller ion radius of Al^{3+} compared to Fe^{3+}, which benefits the small polaron migration and results in improved hematite conductivity and photoresponse.[147]

3.2.3 Electrocatalysts. Electrocatalysts are always used to decrease the onset potential of a photoelectrode. The researchers preferred a photo-assisted electrodeposition method to provide a more uniform distribution of Co-Pi onto hematite made by APCVD. The state of the art hematite photoanode was obtained whose IPCE reached 18% at 440 nm and 1 V_{RHE}, and photocurrent kept stable over 70 h.[29] In terms of overpotential, IrO_2 is the best WOC.[157] The photocurrent of hematite synthesized by APCVD with a surface loading of IrO_2 reached 3.3 mA cm^{-2} under AM 1.5 G of 100 mW cm^{-2}, at 1.23 V_{RHE}.[48] The onset potential after this surface modification shifted from 1.0 to 0.8 V_{RHE}. Surfaces loaded with Co^{2+} and RuO_2 have also been investigated, though those effects are not so impressive.[131,158]

3.2.4 Passivating layer. While water oxidation catalysts can decrease the onset potential, passivating the hematite surface can achieve the same effect. This can be done by depositing Al_2O_3 onto hematite surface,[74] or by just impregnating hematite into a metal salt solution, then heating, and then dissolving to form a hematite–metal oxide solid solution layer to passivate the surface states or reduce the back reaction caused by electrons leaking at the hematite–electrolyte interface.[159] Recently, a Ga_2O_3 underlayer as an isomorphic template for ultrathin hematite films has been found to lead to efficient PEC water splitting.[160] Nb_2O_5 or TiO_2 oxide underlayers also enhance the performance of ultrathin hematite photoanode for water splitting.[161]

3.3 TaON and Ta_3N_5

Beside $BiVO_4$ and hematite oxide photoelectrodes, some (oxy) nitrides especially TaON and Ta_3N_5, emerge as promising photoelectrode candidates for PEC water splitting, based on the fact that they are able to reduce and oxidize water in the presence of appropriate sacrificial reagents under visible-light irradiation in the powder suspension system.[162] Their conduction band minimum and valence band maximum straddle the water oxidization and reduction potentials,[163] which, in principle, is required for PEC water splitting without external bias. In addition, the discrepancies between the conduction band minimum and hydrogen redox potential as well as the valence band maximum and oxygen redox potential provide an additional potential to compensate the required overpotential, which drives their respective half reactions (water reduction and water oxidation reactions).[163] Of the two half reactions, water oxidation is more complex and challenging.[164] TaON and Ta_3N_5 exhibit exciting water oxidation quantum yields of 34% and 10%, respectively,[165,166] which makes them highly efficient unassisted photoelectrode candidates working under visible light irradiation.[162]

3.3.1 Preparation of Ta_3N_5 and TaON photoelectrodes. Owing to the rigorous reaction conditions, the synthesis of Ta_3N_5 and TaON photoelectrodes by nitridation of Ta_2O_5 are restricted to several limited techniques. A simple procedure involves the surface oxidation of Ta foil followed by nitridation of the as-formed Ta_2O_5 layer on Ta foil under NH_3 flow.[167] The drawback of this method is the poor control of the film thickness as well as the structural discontinuities (cracks at the surface and gaps between the product layer and the substrate), which develop from expansion and contraction of the Ta sheet during the oxidation and nitridation. In an attempt to eliminate these drawbacks, radio-frequency magnetron sputtering technology is adopted to control precisely the composition and thickness of a film.[168] Packed Ta_3N_5 film can be directly grown on the substrate without undesirable surface cracks or gaps between the product layer and the substrate. However, the TaON byproduct must be removed, since the structure imperfections may behave as recombination centers for photogenerated electrons and holes, thus lowering the PEC performance. In addition, there are reports on atomic layer deposition for the direct generation of Ta_3N_5 or TaON films, and pulsed laser deposition for the growth of Ta_2O_5 film followed by thermal nitridation under ammonia to produce Ta_3N_5 and TaON films.[169,170]

Apart from the above methods, the direct growth of TaON and Ta_3N_5 or their Ta_2O_5 precursor on substrates can be realized by anodization of Ta foil, thus forming one dimensional Ta_2O_5 nanotube arrays, which are vertically grown on the Ta foil. By controlling the nitridation parameters, vertically oriented one-dimensional TaON and Ta_3N_5 nanotube array films have been successfully synthesized.[171–173] The vertically aligned TaON and Ta_3N_5 nanotube array films provide continuous pathways for electron transportation, and thin walls reduce the hole transport distance, thus facilitating the separation of photogenerated charges. Besides, tube structures offer large internal surface areas and benefit the access of electrolyte, thus better utilizing the semiconductor–electrolyte interfaces. Also, profiting from the reduced hole transport distance, the length of nanotube arrays can be finely tuned, which in turn allows most of the solar light to be captured.[172] However, to preserve the tubular structure, neither the calcination temperature nor the nitridation of Ta_2O_5 nanotube arrays can be sufficiently high, thus making as-prepared TaON or Ta_3N_5 nanotube array films highly defective and less crystalline. For example, TaON nanotube array films show enhanced photon absorption properties up to *ca.* 600 nm, which are quite similar to those of Ta_3N_5, suggesting that a significant amount of defects exist in the TaON nanotube array films.[171] These defects in TaON or Ta_3N_5 photoelectrodes are believed to be deleterious by serving as recombination centers for electron–hole pairs. The elimination of defects in TaON and Ta_3N_5 nanotube array photoelectrodes must resort to suitable modification strategies and/or fine control of the preparation parameters.[173]

The disadvantage existing in the above systems is the nontransparent nature of the employed substrates, which confines the illumination direction, so that only the front side can be illuminated. Correspondingly, TaON or Ta_3N_5 films formed on transparent conducting glass supports harbor great advantages over the nontransparent substrates. Furthermore, the fabrication of TaON and Ta_3N_5 films on conducting glass supports usually involves the deposition of TaON and Ta_3N_5 particles. TaON and Ta_3N_5 particles will thus be prepared in advance, affording opportunities for precise control of the final microstructures of the TaON and Ta_3N_5 films.

The doctor blading technique is the most commonly adopted method for the preparation of thin films fixed on conducting glass substrates. The particle precursors are mixed with organic additives and a small amount of water to form a paste by grinding in mortar, followed by scraping the paste onto a substrate. A controllable thickness of film can be obtained; although the reproducibility is good, the precision of film thickness is in the micron range. TaON and Ta_3N_5 thin films have been achieved through this doctor blading technique.[174,175] The biggest shortcoming of the doctor blading technique is the introduction of a large amount of high-boiling organic species, which can only be eliminated through high temperature calcinations. Unfortunately, high temperature treatment basically leads to disastrous damage to the final properties of the TaON and Ta_3N_5, stemming from their irreversible oxidation or decomposition of surface or bulk during the high temperature process, even under an inert atmosphere. EPD is an alternative method for the deposition of TaON and Ta_3N_5 films on conducting glass substrates, which may introduce fewer organic species.[56] The film thickness can be controlled in the submicron range, and the low boiling solvent can easily be removed by evaporation at low temperatures, thereby avoiding the high temperature calcinations. Meanwhile, EPD is also applicable to nonplanar and even complex multi-dimensional substrates, and enables facile deposition of particles on a variety of substrates.[176] TaON and Ta_3N_5 thin films spread on FTO conducting glass supports have been achieved by EPD with good reproducibility.[58]

3.3.2 PEC performance of Ta_3N_5 and TaON photoelectrodes. Table 3 lists PEC performances of some Ta_3N_5 and TaON photoelectrodes. During the PEC water splitting process, Ta_3N_5 and TaON photoelectrodes have exhibited poor efficiencies (maximum IPCE 5.3% at 450 nm, 0.5 V *vs.* Pt in 1 M KOH)[11] and terribly discouraging photochemical stabilities (persisting for only several minutes) until Abe and coworkers made a substantial contribution to the highly efficient Ta_3N_5 (*ca.* 31% at 500 nm, 1.15 V_{RHE}) and TaON (*ca.* 76% at 400 nm, 1.15 V_{RHE}) photoelectrodes by using the EPD method.[58,172] They have highlighted the importance of efficient contacts among the particles, because efficient electron transport in the porous electrodes will be ensured by these bridge-like contacts.[58,177] The rigid surfaces formed at high temperatures and the refractory characteristics of Ta_3N_5 or TaON materials need high energy input to promote the sintering and necking of Ta_3N_5 or TaON particles. However, owing to the temperature limit of conducting glass substrates and the bad antioxygenic properties of Ta_3N_5 or TaON, effective contacts among Ta_3N_5 or TaON particles cannot be successfully formed by simple heat treatment. The isolated particles greatly block the electron transport, which results in low PEC efficiencies. Necking treatment is therefore indispensable for forming interconnected particles film.[177,178] $TiCl_4$ post-treatment for forming TiO_2 joints among the TaON and $LaTiO_2N$ particles is reported to improve evidently the photocurrent, while the post-calcination under an inert atmosphere is crucial to avoid the oxidation of oxynitrides and to transform $TiCl_4$ to TiO_2.[174,179] In this case, TiO_2 bridges among isolated particles behave as electron transfer media, which is favorable for the improvement of photocurrent, but the photocurrent benefits may be restricted by the hetero-junction interfaces between TiO_2 and these oxynitride particles. Ta_2O_5 bridges may also fail in the same way in obtaining desirable PEC properties on TaON or Ta_3N_5 photoelectrodes. Upon heat treatment under NH_3 atmosphere, TaON bridges formed between TaON or Ta_3N_5 particles by nitriding the amorphous Ta_2O_5 bridges from $TaCl_5$, producing a considerable increase in the photocurrent of TaON or Ta_3N_5 photoelectrodes. For TaON photoelectrodes, the transformation of bridges from Ta_2O_5 to TaON enables better electron transport pathways by forming more conductive TaON bridges and interconnected single

Table 3 PEC performances of Ta_3N_5 and TaON photoelectrodes

Photoelectrode	Preparation	Catalyst	Electrolyte	IPCE or Photocurrent	Ref.
Ta_3N_5	Reactive sputtering	IrO_2	0.1 M Na_2SO_4, pH 8.5	*ca.* 2.3 mA cm^{-2} under irradiation of 300 W Xe lamp with cutoff $\lambda \geq 420$ nm	168
Ta_3N_5	Oxidation and nitridation of Ta foil	—	0.1 M K_2SO_4 adding KOH, pH 11	*ca.* 1% at 420–550 nm and −0.1 V *vs.* Ag/AgCl (0.75 V_{RHE})	167
TaON	EPD	IrO_2	0.1 M Na_2SO_4, pH 6	*ca.* 76% at 400 nm at 0.6 V *vs.* Ag/AgCl (1.15 V_{RHE})	56
Ta_3N_5	EPD	IrO_2	0.1 M Na_2SO_4, pH 6	*ca.* 31% at 500 nm and 1.15 V_{RHE}	58
Ta_3N_5	EPD	Co_3O_4	1 M NaOH	26% at 400 nm and 24% at 550 nm (1.2 V_{RHE}); 40% at 400 nm and 36% at 550 nm (1.4 V_{RHE})	57
TaON	Anodization and nitridation of Ta foil	—	1 M KOH	2.6 mA cm^{-2} at 1.52 V_{RHE} (AM 1.5)	171
Ta_3N_5	Anodization and nitridation of Ta foil	—	1 M KOH	5.3% at 450 nm and 0.5 V *vs.* Pt pseudo-reference	172
TaON	TaON powder spreading	—	0.1 M Na_2SO_4	*ca.* 0.023 mA cm^{-2} at 1.0 V *vs.* Pt pseudo-reference under irradiation of 300 W Xe lamp with cutoff $\lambda \geq 420$ nm in two-electrode configuration	174
Ta_3N_5	Ta_3N_5 powder spreading	—	0.1 M Na_2SO_4	*ca.* 0.10 mA cm^{-2} at 1.55 V_{RHE} (300 W Xe lamp with cutoff $\lambda \geq 420$ nm)	175
$Ta_yCo_{1-y}N_x$	Dip coating and nitridation	Co oxide	0.1 M Na_2SO_4 adding NaOH, pH 11	25% at 450 nm and 1.45 V_{RHE}	180
TaON	EPD	CoO_x	Sodium phosphate buffer solution, pH 8	*ca.* 42% at 400 nm and 1.2 V_{RHE}	59
Ta_3N_5	Anodization and nitridation of Ta foil	IrO_2	0.1 M Na_2SO_4 adding KOH, pH 11	IPCE ∼10% at 400 nm and 1.45 V_{RHE}	181
Ta_3N_5	Evaporation of Ta metal followed by nitridation	Co^{2+}	1 M KOH, pH 13.6	IPCE ∼9% at 500 nm and 1.2 V_{RHE}	182

phase TaON film.[177] Compared with TaON photoanodes, the inferior performance on Ta_3N_5 photoanodes may be associated with TaON bridges, in which relatively low electron transport may occur across the hetero-interfaces originating from the crystal structure discrepancy between TaON and Ta_3N_5. Transparent conducting oxide substrates which can withstand higher temperatures are desirable for further photocurrent improvement of Ta_3N_5 photoelectrodes by the formation of Ta_3N_5 bridges.

The concentration and lifetime of carriers also play pivotal roles in PEC performance, and increasing the carrier densities and conductivities of Ta_3N_5 and TaON should be conducted to facilitate the separation and transfer of charge carriers. Doping of Ta_3N_5 and TaON with appropriate ions may be a good way to obtain Ta_3N_5 and TaON materials with suitable conductivities and carrier densities. The electronic structures of TaON and Ta_3N_5 after doping may be varied, and formation of impurity levels (for example Ta^{4+}) may simultaneously be suppressed.

Apart from meliorating the electronic properties of TaON and Ta_3N_5, reducing the feature sizes of Ta_3N_5 and TaON by nanostructure may greatly benefit the separation of electron–hole pairs and promote hole consumption by water, through lessening the distances that minority carriers have to travel to the reactive surfaces and increasing the semiconductor–electrolyte interface areas.[48]

3.3.3 Photochemical stability of Ta_3N_5 and TaON photoelectrodes. Although current Ta_3N_5 and TaON photoelectrodes can exhibit high efficiencies for PEC water splitting, their photocurrent densities often sharply decrease over the illumination time. Improvement in their photochemical stabilities is therefore longstanding and highly urgent for their practical utilization. Their poor photochemical stabilities are derived from the self-oxidative decomposition of TaON and Ta_3N_5 materials, in which N anions in lattice are oxidized to form N_2 by photogenerated holes. As long as the consumption of photogenerated holes by water oxidation is inefficient, hole accumulation will be alternatively alleviated through self-oxidative decomposition of TaON and Ta_3N_5 materials, which imposes irreversible damage to these photoelectrodes. Therefore, the photochemical stabilities of TaON and Ta_3N_5 photoanodes are intrinsically determined by their manner of hole consumption; efficient consumption of holes by water oxidation, as opposed to self-oxidative decomposition, which undoubtedly resorts to accelerated kinetics of water oxidation.

The selection of WOCs and deposition method are considered to be effective and straightforward means to improve the photochemical stabilities of Ta_3N_5 and TaON photoelectrodes, for WOCs deposited onto the TaON and Ta_3N_5 surface bear the responsibility of capturing the holes and reacting with water to evolve O_2. Although IrO_2 has been reported as the best WOC and functions efficiently independent of pH for electrochemical water oxidation,[183] it performs dissatisfactorily towards long-term stable Ta_3N_5 and TaON photoelectrodes, which may derive from the loose adhesion and poor distribution of IrO_2 on TaON and Ta_3N_5 photoelectrode surfaces.[56] Further advances in their photochemical stabilities may come from the introduction of other promising WOCs and ingenious loading strategies. The nature of the WOC–photoelectrode junction should also be taken in consideration, for the junction acts as the pathway for charge carriers.[184]

One also notes that not only the photocatalytic activities for O_2 evolution but also the durabilities of TaON and Ta_3N_5 are terribly disappointing in acidic solution.[185] Only with the assistance of La_2O_3, which behaves as a pH buffer, can TaON and Ta_3N_5 achieve steady and prompt O_2 production.[185] Although IrO_2 is insensitive to the electrolyte environment, the large portions of uncovered TaON and Ta_3N_5 surfaces after IrO_2 modification may suffer from an acidic environment, for the production of O_2 can decrease the pH at the semiconductor–electrolyte interface. Water oxidation involves the elimination of protons, and since a low proton removing rate may contribute to water oxidation kinetics, special attention should also be paid to the effects of electrolytes.[186–188] Buffered solutions in the alkaline range and basic electrolytes may hold great advantages over the presently used Na_2SO_4 electrolyte (slightly acidic) for the stabilization of TaON and Ta_3N_5 photoanodes, on account of their excellent proton accepting abilities. Several studies speciously suggest that Ta_3N_5 exhibits a higher PEC performance in alkaline solution than in a neutral pH electrolyte.[167,171,172]

Cobalt-phosphate and cobalt-borate WOCs working in a neutral or slightly basic environment may show their superiority for improving the photochemical stabilities of Ta_3N_5 and TaON photoelectrodes, due to their robust water oxidation properties and uniform distribution on photoelectrodes, resulting from facile electrodeposition.[188]

The present authors have found that the PEC durability of Ta_3N_5 photoelectrodes is improved significantly by loading nanoparticulate Co_3O_4 WOCs, which are uniformly distributed onto the Ta_3N_5 surface and form compact Co_3O_4–Ta_3N_5 junctions. Profiting from the unique features of the present Co_3O_4 WOCs and alkaline solution, the improved kinetics of water oxidation gives the Co_3O_4/Ta_3N_5 photoanodes the best durability against photocorrosion to date (shown in Fig. 4).[57] Abe and coworkers have found that CoO_x WOCs obviously enhance the PEC stability of TaON photoelectrodes.[59] They have also emphasized that slightly basic sodium phosphate buffer solution (pH 8) is indispensible for high PEC stability of TaON photoelectrodes.[59]

Meanwhile, the highly defective natures of Ta_3N_5 and TaON, which are ascribed to the formation of reduced tantalum species during the nitridation process,[189,190] contribute greatly to their suppressed PEC performances and stabilities. As a result, highly-crystalline and less-defective Ta_3N_5 and TaON photoelectrodes are highly desirable, which count on precise control of preparation parameters. Appropriate surface modification may also contribute to less-defective Ta_3N_5 and TaON photoelectrode materials.[173]

To summarise, accelerating the kinetics of water oxidation by application of robust WOCs, melioration of material properties, deposition of a surface protection layer, and nanostructuring of semiconductors will substantially boost the practical application of unassisted sunlight-driven H_2 production by TaON and Ta_3N_5 photoelectrodes. We believe the full potentials of TaON and Ta_3N_5 photoanodes will be implemented by continued parallel optimization of these influencing factors.

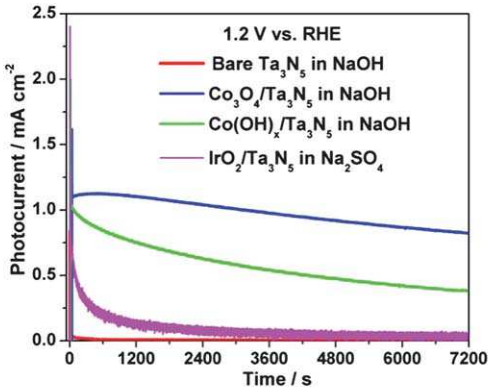

Fig. 4 Photocurrent decay curves measured at 1.2 V_{RHE} under visible light irradiation ($\lambda \geq 420$ nm) for Co_3O_4/Ta_3N_5 (in blue), $Co(OH)_x/Ta_3N_5$ (in green), bare Ta_3N_5 (in red) photoelectrodes in 1 M NaOH (pH = 13.6) solution and IrO_2/Ta_3N_5 (in mauve) in 0.5 M Na_2SO_4 solution (pH = 6.5). The $Co(OH)_x/Ta_3N_5$ photoanode was prepared by impregnation of $Co(OH)_x$ colloids onto Ta_3N_5, the IrO_2/Ta_3N_5 photoanode was prepared by impregnation of IrO_2 colloids onto Ta_3N_5, and the Co_3O_4/Ta_3N_5 photoanode was prepared by calcining $Co(OH)_x/Ta_3N_5$ at 573 K for 10 minutes in air.[57]

4 Methods to improve PEC efficiency

4.1 Tuning the band structure of a photoelectrode by doping with elements

Tuning the band structure of a photoelectrode by doping it with elements is one of the most common methods to improve the PEC efficiency. In the case of a wide bandgap photoelectrode, for example TiO_2, tuning its band structure by doping with elements (such as N, S, C and transition metals) is often applied to narrow the bandgap and make the light absorption edge red-shift, which in turn raises the PEC efficiency. However, for the wide bandgap photoelectrode doped with elements, the absorption coefficients in the visible region are very low. Also, the doping with elements brings crystal defects and may form deep level defects, which may increase the recombination probability of photoinduced electron and hole pairs, thus reducing the IPCE in the ultraviolet region. As a result, only a limited increase in STH may be obtained for wide bandgap photoelectrodes, though the action spectra of doped photoelectrode may red shift because of the narrowing bandgap.[14,15,18,19] Codoping with two nonisovalent elements is also used to balance the charge of the wide bandgap photoelectrode. For example, a synergistic effect involving hydrogenation and nitridation cotreatment of TiO_2 nanowire arrays (codoping with H^+ and N^{3-}) can improve the water photooxidation performance under visible light illumination.[191]

In order to improve significantly the STH of a PEC cell, care must be taken in choosing photoelectrode materials with narrow intrinsic bandgaps. In the case of photoelectrodes with narrow bandgaps, such as Fe_2O_3 and $BiVO_4$, tuning their band structures by doping with nonisovalent elements is often used to increase the carrier concentration and the minority carrier diffusion length.[17,30,55,120] Obviously, the decrease in the effective mass of the minority-carrier is helpful for increasing the minority carrier diffusion length, thus improving the photocurrent density.

4.2 Surface treatment

Surface treatment is another efficient method to improve the PEC efficiency of a photoelectrode by removing or diminishing the segregation near surface or surface states, which often act as recombination centers of photoinduced carriers.

Segregation, referring to the enrichment of a material constituent, often emerges at a free surface or an internal interface of a material during the preparation. The segregation in the photoelectrodes will influence their PEC properties. For example, we have reported that indium-rich segregation in the $In_{0.20}Ga_{0.80}N$ photoelectrode can act as surface recombination centers of photoinduced electrons and holes.[192] After the electrochemical surface treatment (electrochemically corroded in 1 M HBr aqueous solution), the $In_{0.20}Ga_{0.80}N$ photoelectrode's IPCE at 400 nm is increased by a factor of three (as shown in Fig. 5a). A photoluminescence peak around 560 nm disappears (as shown in Fig. 5b), suggesting that indium-rich segregation is removed by this electrochemical surface treatment. In addition, the photocurrent can be kept constant after a long-time illumination, which is beneficial for practical application in solar hydrogen production.

A similar method is applied to increase the IPCE of Mo-doped $BiVO_4$ photoanodes by the present authors.[193] The photocurrent of the Mo-doped $BiVO_4$ photoanodes under visible-light illumination from the front side can be improved significantly after electrochemical surface pretreatment, due to the dissolution of the MoO_x segregation phase at high potential and under illumination. Photoetching in 1 M Na_2SO_3 has been early used to enhance the IPCE of CdS photoelectrodes, because photoetching can remove from the electrode surface a thin layer rich in oxygen and sulphur vacancies where recombination is favored.[71] Very recently, the present authors have reported a facile strategy to passivate surface states (diminishes the recombination centers of the photogenerated carriers) on the undoped hematite photoanode by scanning the photocurrent–time curves in saturated NaCl aqueous electrolyte, thus leading to an increase in the hematite IPCE by a factor of 2.6 (from 5% to 13%) at 365 nm and 1.23 V_{RHE}.[75]

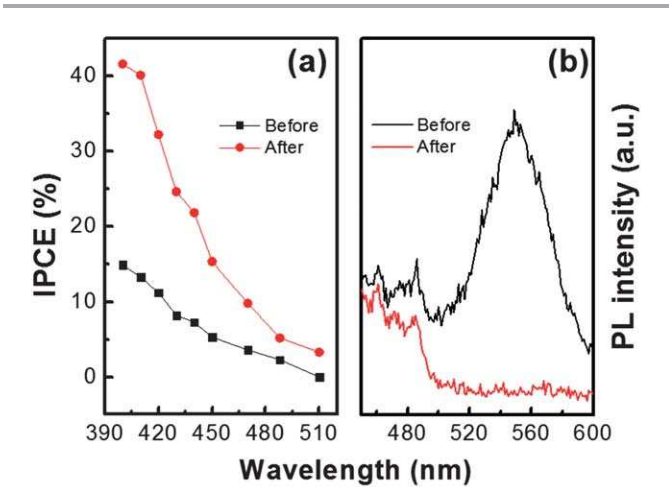

Fig. 5 (a) Action spectra of $In_{0.20}Ga_{0.80}N$ photoelectrode in 1 M HBr solution before and after electrochemical corrosion, the potential is 0.8 V *vs.* Ag/AgCl. (b) Photoluminescence properties of $In_{0.20}Ga_{0.80}N$ before and after the surface treatment of electrochemical corrosion.[192]

4.3 Electrocatalysts

For the reaction of PEC water splitting, an overpotential is required to overcome the kinetic limitation and drive the desired chemical reaction. Electrocatalysts can be divided into WOC and WRC (water reduction catalyst).[5,6] Electrocatalysts (or named catalysts) act as activation sites for the evolution of hydrogen (photocathode) or oxygen (photoanode) and improve the kinetics of the reaction, thereby reducing the bias voltage and enhancing IPCE at low bias. However, it is difficult to increase the plateau photocurrent of a photoelectrode. A good electrocatalyst must match two basic requirements. First, the electrocatalyst should exhibit a minimum overpotential for water reduction/oxidation reaction, and hydrogen or oxygen can evolve as quickly as the semiconductor photoelectrodes can supply photoinduced electrons or holes to the electrocatalyst.[5] Second, an effective electrocatalyst must show long-term chemical and photochemical stability.[5]

Among the WRCs employed so far, noble metals like Pt, Ru, and Pd represent the most common ones loaded onto the surface of photocathodes for promoting the evolution of hydrogen.[194–196] Besides noble-metal WRCs, promising candidates include MoS_2, Mo_3S_4, W–Ni–P, W–Cu, and Ni–Fe.[197–200] WOCs, such as IrO_2, Co_3O_4 and amorphous cobalt-phosphate (Co-Pi), are usually applied to enhance the efficiency and reaction kinetics of photoanodes.[183,186,201] "Co-Pi" is one attractive electrocatalyst for PEC water splitting recently though its precise microscopic identity remains uncertain.[188,202] It has been used in various photoelectrodes, including ZnO, WO_3, silicon, and hematite.[187,203–206]

4.4 Morphology control of photoelectrodes

To obtain high efficiency, a photoelectrode film must be thick enough to absorb more light, and photogenerated carriers deep within the photoelectrode must be able to reach the surface, where they can be collected.[5] Therefore, the minority carrier diffusion length is vital for a PEC cell. There are some cases for α^{-1} (penetration depth of light) as following:

(i) If $\alpha^{-1} > L_{SC} + L_p$ (effective transfer distance of minority carriers, shown in Fig. 2) in a photoelectrode, the photoinduced minority carriers in $L_{SC} + L_p < x < \alpha^{-1}$ will be recombined during the diffusion process, while the photoinduced minority carriers in $0 < x < L_{SC} + L_p$ may reach the surface by diffusion and/or drift.

(ii) If $\alpha^{-1} < L_{SC}$, near all of the photoinduced minority carriers in $0 < x < \alpha^{-1}$ may reach the surface by drift.

(iii) If $L_{SC} < \alpha^{-1} < L_{SC} + L_p$, the photoinduced minority carriers in $0 < x < \alpha^{-1}$ may reach the surface by diffusion and drift.

The morphology of a photoelectrode will influence the transfer distance of photoinduced carriers. Morphology control of photoelectrodes, in particular those with porous structures or nanostructures (such as nanorods, nanotubes, and nanowires), has attracted much attention, since the potential advantages of nanostructured photoelectrodes over conventional semiconductor photoelectrodes are the relatively short transfer distance of minority carriers and the increase in the surface reaction sites, which is helpful for lowering the recombination probability of carriers and improving their PEC performances.[171,207–210]

Three-dimensional nanowire networks are promising for designing high-performance photoelectrodes owing to their long optical path for efficient light absorption, high-quality one-dimensional conducting channels for rapid electron hole separation and charge transportation, as well as high surface areas for fast interfacial charge transfer and electrochemical reactions.[211] Recently, hierarchical structures of photoelectrodes have attracted interest. Wang *et al.* have reported that in the TiO_2–Si three-dimensional nanowire architecture, Si nanowires act as conductive paths for fast electron transportation, while water oxidation occurs on the TiO_2 nanowire surfaces.[212] Three-dimensional hierarchical nanostructures have been synthesized by the halide chemical vapor deposition of InGaN nanowires on Si wire arrays.[213] The photocurrent of hierarchical Si/InGaN nanowire arrays increased by 5 times, compared with that of InGaN nanowire arrays grown on planar Si (1.23 V_{RHE}).[213]

4.5 Other methods

Heterojunction photoelectrodes, which often consist of a narrow bandgap semiconductor photoelectrode covered by a wide bandgap one, have been also used for improving not only IPCE by enhancing the light absorption of the photoelectrode,[214] but also its photochemical stability by avoiding contact with narrow bandgap semiconductors and electrolytes. In general, the PEC properties of heterojunction photoelectrodes are strongly determined by the outer photoelectrode.[64] Deposition of an In_2O_3 overlayer can multiply the photocurrents of $LaTiO_2N$ photoanodes by a factor of 2–3, which could be attributed to better charge separation and lower charge recombination and/or to better charge extraction by the action of a surface dipole.[215] There are also reports on heterojunction photoelectrodes consisting of p- and n-type hematite semiconductors. For example, Wang *et al.* reported that the p-type hematite layer with a thickness of 5 nm grown on n-type hematite can obtain a nominal 200 mV turn-on voltage shift toward the cathodic direction, which favors PEC water splitting reactions.[128]

Recently, some new methods have been developed to enhance the PEC performances of photoelectrodes. Some oxide layers (such as Al_2O_3 and Ga_2O_3) loading onto the surface of hematite can also improve the PEC performances because they passivate the surface states or reduce back-reactions caused by electrons leaking at the hematite–electrolyte interface.[74,159] Yang *et al.* have observed an enhanced photocurrent in a thin-film iron oxide photoanode coated on arrays of Au nanopillars, due primarily to the increased optical absorption originating from both surface plasmon resonances and photonic-mode light trapping in the nanostructured topography.[216] Wang *et al.* have demonstrated a strain-related photocurrent variation using a ZnO-based piezotronic-photoelectrochemical anode, and the photocurrent increased/decreased by a fixed amount when the piezotronic-photoelectrochemical anode was under a constant tensile/compressive strain, respectively.[217] Li *et al.* have reported the design and characterization of a novel double-sided CdS and CdSe quantum dot cosensitized ZnO nanowire array photoanode for PEC hydrogen generation.[218]

In addition, photoelectrodes of photonic crystals have been introduced into PEC cells.[219,220] The light in a photonic crystal undergoes strong coherent multiple scattering and travels with very low group velocity near the stop band edges of semiconductor photoelectrodes. This slow-light effect can considerably increase the effective optical path length, therefore leading to a delay and storage of light in the photoelectrodes, which in turn enhances the IPCE, for example WO_3.[219]

5 Photochemical durability of photoelectrodes

An essential requirement for the photoelectrodes is resistance to reactions such as electrochemical corrosion, photo-corrosion and dissolution at the solid–liquid interface, which result in degradation of their PEC properties.[23]

5.1 Thermodynamics views

Aside from PEC activities, the stability and long-term performance of a photoelectrode have been primary concerns for the development of PEC hydrogen production. The chemical and photochemical corrosion of photoelectrodes must be overcome for application. The photoinduced holes and electrons in the photoelectrodes are often characterized by strong oxidizing and reducing potentials, respectively. Besides being injected into the electrolyte to drive redox reactions, these holes and electrons may oxidize or reduce the photoelectrodes themselves, thus leading to the decomposition.[221] A Pourbaix diagram (also known as a potential/pH diagram) is used to analyse corrosion of materials, because it maps out the possible stable phases of an aqueous electrochemical system.[222] Photochemical corrosion can be described by the model presented by Gerisher and Bard.[223,224] The photochemical corrosion of an MX semiconductor, which leads to anodic and cathodic decomposition, may be represented by the following reactions, respectively,

$$MX + ne^- \rightarrow M + X^{n-}\ (\varphi^0_{dec,n}) \quad (9)$$

$$MX + nh^+ \rightarrow M^{n+} + X\ (\varphi^0_{dec,p}) \quad (10)$$

where n is the number of electrons or holes; and $\varphi^0_{dec,n}$ and $\varphi^0_{dec,p}$ are the free energy change at the cathode and anode, respectively. In the case of standard hydrogen electrode, $F_{dec,n}$ and $F_{dec,p}$ is the potential corresponding to redox and oxide reaction of the photoelectrode, respectively. The chemical stability of a photoelectrode in view of thermodynamics is shown in Fig. 6.

The following criteria for the stabilities of photoanodes or photocathodes against electrochemical corrosion have been formulated by Gerisher,

$$E_{O_2/H_2O} < F_{dec,p} \quad (11)$$

$$E_{H^+/H_2} > F_{dec,n} \quad (12)$$

where E_{H^+/H_2} and E_{O_2/H_2O} are the energy of the redox couple H^+/H_2 and O_2/H_2O, respectively.[224]

For improving PEC stability of a photoelectrode, a proper electrolyte should be selected. For example, the present authors have studied the effects of different electrolytes with the same pH value (0.1) on the photochemical stability of $In_{0.20}Ga_{0.80}N$ photoanodes under visible-light irradiation. The results show that $In_{0.20}Ga_{0.80}N$ photoanode has better PEC stability in HBr solution, compared with in HCl or H_2SO_4 solution.[225] We can conclude that $E_{Br^-/Br_2} < F_{dem,p}$ (InGaN) $< E_{O_2/H_2O}$.

5.2 Kinetic views

Accelerating the kinetics of water oxidation should be emphasized for improving PEC stability of photoanodes, as the slow kinetics of water oxidation often leads to hole accumulation.[57] For metal oxide photoanodes, the hole accumulation usually results in the recombination of the photogenerated electrons and holes, which is unfavorable for water splitting, but can be effectively circumvented by large applied potentials. When it comes to (oxy)nitride photoanodes, such as Ta_3N_5 and TaON, hole accumulation will be alternatively alleviated through irreversible and self-oxidative decomposition of (oxy)nitride photoanodes themselves, as long as the consumption of photogenerated holes by water oxidation is inefficient. In this connection, the PEC stability of (oxy)nitride photoanodes, such as TaON and Ta_3N_5, are intrinsically determined by the manner of hole consumption. The efficient consumption of the photogenerated holes by water oxidation other than self-oxidative decomposition undoubtedly resorts to acceleration of the kinetics of water oxidation. By employing Co_3O_4 WOC modified Ta_3N_5 photoelectrodes as well as an alkaline electrolyte, the prominent improvement in photostability during water splitting has been experimentally demonstrated by the present authors.[57]

5.3 Chemical stability of photoelectrodes and electrocatalysts

In a PEC cell, not only corrosion of the photoelectrodes, but also corrosion of the electrocatalysts should be considered. For example, RuO_2 and IrO_2 are more stable in acid, while anodic dissolution occurs in base. Co_3O_4 and Co-Pi are more stable in base than acid.

Recently, Li *et al.* reported enhancement in the photochemical stability of hydrogen-treated WO_3 nanoflake.[226] It is ascribed to the formation of substoichiometric WO_{3-x}, which is highly resistant to re-oxidation and peroxo-species induced dissolution.[226] Wang *et al.* have reported water splitting by WO_3 prepared by ALD and decorated with an oxygen-evolving Mn complex catalyst.[227] When the Mn complex catalyst is interfaced with WO_3, it acts as a protecting layer without adverse effect on PEC water splitting, and therefore WO_3 photoelectrodes can be stable in neutral solution.[227]

6 Solar seawater splitting

Solar water splitting is a clean and renewable method of hydrogen production. There is limited work on solar seawater splitting.[228–231] Different from pure water, plenty of Cl^- ions exist in seawater and are oxidized more easily than H_2O oxidation in kinetics due to the lower overpotential. Generally, in seawater splitting, two reaction processes may happen:

$$\text{Photoanode reaction: } 2Cl^- + 2h^+ \rightarrow Cl_2 \quad (13)$$

$$\text{Cathode reaction: } 2H^+ + 2e^- \rightarrow H_2 \quad (14)$$

However, in the experiments, oxygen evolution can be detected, perhaps owing to water splitting or the following reactions.[55]

$$Cl_2 + H_2O \rightarrow H^+ + Cl^- + HClO \quad (15)$$

$$HClO \rightarrow H^+ + Cl^- + 1/2O_2\uparrow \quad (16)$$

A part of the Cl_2 dissolved in the seawater reacts with H_2O, thus forming HClO. Oxygen may come from succeeding photodissociation of HClO.

Note that the Mg^{2+} and Ca^{2+} ions will influence the photochemical stability of the photoelectrodes. In NaCl aqueous

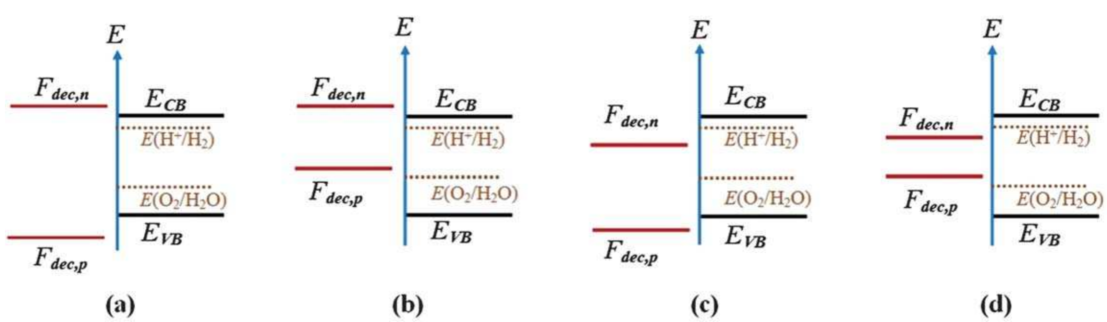

Fig. 6 Chemical stability of a photoelectrode without any bias in view of thermodynamics (a) stable, (b) anodic corrsion, (c) cathodic corrsion, (d) unstable.

solution without Mg^{2+} and Ca^{2+} the pH value of the electrolyte will increase after H^+ is reduced to H_2, which may corrode some photoelectrodes. Mg^{2+} and Ca^{2+} existing the seawater can react with OH^- to form white $Mg(OH)_2$ and $Ca(OH)_2$ precipitates, which keeps the pH of seawater near the photoelectrodes neutral. Hematite and WO_3 photoanodes exhibit good photochemical stability only in base or acid aqueous solution, respectively. Therefore, in seawater splitting, $BiVO_4$ photoanodes exhibit a much better photochemical stability than hematite and WO_3 photoanodes. In addition, the production ratio of O_2 to Cl_2 can be tuned by applying a suitable electrocatalyst, thus obtaining selectivity.

7 Summary and perspectives

In summary, PEC water splitting remains one of the "holy grail" technologies for hydrogen production without producing additional CO_2 in the atmosphere. To date, some photoelectrodes such as $BiVO_4$, α-Fe_2O_3, TaON and Ta_3N_5 have exhibited good PEC efficiencies.

The $BiVO_4$ photocurrent has reached 3.3 mA cm^{-2} under AM 1.5 G irradiation and 1.6 V_{RHE} (STH = 4.1%),[55] which is near half of the theoretical value (7.4 mA cm^{-2}). $BiVO_4$ photoelectrodes also exhibit good PEC durability of over 100 hours under AM 1.5 G irradiation in our lab. Electrocatalyst and microstructure control (porous structure) will be used to improve the PEC performances of the $BiVO_4$ photoelectrodes. More attention should be paid to PEC seawater splitting of $BiVO_4$. $BiVO_4$, as the external layer, may be used to improve PEC durability of heterojunction photoelectrodes.

The hematite photocurrent has reached 3.3 mA cm^{-2} under AM 1.5 G irradiation and 1.43 V_{RHE} (STH = 4.1%),[29] while it is still far away from the theoretical value.[32] As pointed out by Li *et al.*,[126] doping and the control of surface morphology have no major effect on the ultrafast dynamics of the charge carries on picosecond time scales. Poor light absorption is always listed as one of the drawbacks of hematite photoelectrodes.[232] Actually, the absorption coefficient for hematite at 350–400 nm is on the order of 10^5 cm^{-1},[233] which is comparable to that of silicon. The absorption coefficient of hematite decreases rapidly in the visible region, resembling the shape of the action spectrum, as mentioned before. Therefore, the strategies to ameliorate the PEC performances of hematite should be focused on amending its electronic structure or the reducing the density of the trap states, which in turn can lead to a longer photoexcited charge carrier lifetime and more efficient charge separation, compared to the results obtained by altering the nanostructures of hematite.

The IPCE of the TaON photoanodes is *ca.* 76% at 400 nm and 1.15 V_{RHE} in aqueous Na_2SO_4 solution. The plateau photocurrent of Ta_3N_5 has reached ~8 mA cm^{-2} under AM 1.5 G irradiation, which is close to the theoretical value (12.9 mA cm^{-2}). However, the PEC durability of TaON and Ta_3N_5 are only on the order of several hours. The durability of these photoelectrodes may be improved by optimizing the photoelectrode, electrocatalyst, and electrolyte at the same time. In short, accelerating the kinetics of water oxidation by deposition of robust WOCs is an effective way to stabilize TaON and Ta_3N_5 photoanodes. A surface protecting layer, prepared by ALD, is expected to improve the stability of these photoelectrodes. Intensive effort must be made in these areas to achieve substantial advances in the photostability of TaON and Ta_3N_5 photoanodes.

We will make a brief summary of the factors that influence the PEC performances of a photoelectrode. A photoelectrode's electronic structure is the most important factor in determining its PEC properties. Another important factor is the crystallinity of the photoelectrode, which is involved in preparation approaches. The good crystallinity of the photoelectrode favors charge transport, thus improving the PEC performances. Doping with elements is often used to improve the PEC performances of photoelectrodes, for enhancing visible light absorption in wide bandgap photoelectrodes or promoting carrier transport in narrow bandgap photoelectrodes, respectively. The morphology of the photoelectrode films plays a considerable role in influencing the PEC performances. Especially, photoelectrodes with nanostructure (such as nanorod, nanotube, and mesoporous structure) and hierarchical structure exhibit some advantages such as shortening the transfer length, increasing the surface area, and enhancing light absorption. Surface treatment and electrocatalysts are very effective ways not only to enhance IPCE and STH but also improve the photochemical stability. Heterojunction photoelectrodes are also expected to advance the PEC performance.

Fundamental photoelectrode materials and materials-interface properties hold the keys to successful PEC research and development. Breakthroughs of PEC cells are required on several fronts:

(1) To develop the photoelectrode materials with good absorption and carrier-transport properties.

Future developments may involve expansion of the range of materials to include mixed oxide or (oxy)nitride materials with visible-light response, as it has been suggested that modification of charge mobility may raise PEC performances. Accurate control over the composition and the defect chemistry should be required during the development of new photoelectrodes. New photoelectrode materials with a bandgap of *ca.* 2.0 eV, good stability, and long minority carrier diffusion length are required in the future. A high-throughput method has been developed using a commercial piezoelectric inkjet printer for the synthesis and characterization of mixed-metal oxide photoelectrodes for water splitting,[234] which will favor the development of new photoelectrode materials.

(2) To construct PEC devices based on multijunction semiconductors.

The photocurrent densities of $BiVO_4$, α-Fe_2O_3, TaON and Ta_3N_5 photoelectrodes look encouraging, especially for Ta_3N_5, but none of them exhibit enough photopotential under sunlight to split water. To date, the only demonstrations of single-photoelectrode PEC water splitting have utilized very high bandgap materials, such as $SrTiO_3$ and $KTaO_3$.[235,236] PEC devices based on multijunction semiconductors should be paid more attention. In theory, p–n PEC cells consisting of a photoanode and photocathode favor the transport and separate of photoinduced electrons and holes.[237] The p–n PEC structure also relaxes the

PEC stability requirements of the light absorbers, facilitating the use of photocathodes that are stable under cathodic (but not necessarily anodic) conditions, in combination with photoanodes that are stable under anodic (but not necessarily cathodic) conditions.[41] There are some examples of p–n PEC cells, such as p-$SrTiO_3$:Rh/n-$BiVO_4$ and dye-sensitized p-NiO/n-$BiVO_4$, which can break apart water molecules into H_2 and O_2 under visible light irradiation without applying an external bias.[238,239] However, the efficiencies of these p–n PEC cells are often very low, owing not only to high internal resistance but also to the low efficiency of a single photoelectrode. So far p-type semiconductors prepared by traditional ways are very limited. A breakthrough in PEC cells is expected if high efficient p-type photocathodes can be realised. The development of high efficiency and stable p-type photocathodes, such as Cu_2ZnSnS_4 and $Cu_2ZnSn(S,Se)_4$, will push this field forward.[240,241]

The tandem PEC cell is also one way to overcome unfavorable band energetics associated with many photoanodic materials.[30,242,243] Such tandem designs, in principle, allow the use of relatively small band-gap (*i.e.*, 1.1–1.7 eV) light absorbers that are well-matched to the solar spectrum while simultaneously providing the necessary overall photovoltage (>1.23 V) required under standard conditions to produce H_2 and O_2 from H_2O.[41] p-Si is an attractive photocathode candidate for a tandem PEC cell, since Si with a suitable bandgap is earth-abundant and relatively cathodically stable.[41] Hematite has been grown on vertically aligned Si nanowires using ALD to form a dual-absorber system, and Si nanowires absorb photons that are transparent to hematite (600 nm < λ < 1100 nm) and convert the energy into additional photovoltage to assist PEC water splitting by hematite.[244]

(3) To design surfaces or interfaces with energetic and kinetic properties favoring water-splitting reactions, and inhibiting corrosion reactions.[73,79]

Thimsen and coworkers has reported a highly active photocathode for solar H_2 production, consisting of electrodeposited cuprous oxide, which was protected against photocathodic decomposition in water by nanolayers of Al-doped ZnO and TiO_2 and activated for hydrogen evolution with electrodeposited Pt nanoparticles.[245] It has been recently reported that surface protection of unstable Si photoanodes by a thin layer of atomic layer deposited TiO_2 shows great resistance to corrosion in harsh pH and oxidative environments.[246] This emerges as a path breaking technique for stabilizing these unstable semiconductors, which possess huge potentials as photoelectrodes themselves.

Although many impressive results have been achieved, there is still a long way for PEC cells to go into practical use. Innovation and developments in new photoelectrode materials, structure and conceptions are still required.

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